

Master's Thesis

**Last in Translation? Evaluating Language-Invariant
Reason Ranking in Numerical Fact-Checking**

Master's Thesis for a degree in Applied Computer Science (M.Sc.)
Chair for Scientific Information Analytics
University of Göttingen

Submission date: September 24th, 2026

By: Ahmad Raza Khawaja

From: Göttingen

Matriculation number: 25848862

First examiner: Prof. Dr. Bela Gipp
Second examiner: Dr. Jan Philip Wahle
Supervisor: Shakib Yazdani

University of Göttingen
Chair for Scientific Information Analytics

CONTENTS

1	Introduction	1
1.1	Motivation	1
1.2	Problem Statement and Research Gap	2
1.3	Grounding: CLEF CheckThat! 2026 Task 2	2
1.4	Research Questions	3
1.5	Contributions	3
1.6	Thesis Outline	4
2	Background	4
2.1	Large Language Models	4
2.2	Adapting Language Models to a Task	5
2.3	Text Ranking with Language Models	5
2.4	Classification and Training Objectives	7
2.5	Automated Fact-Checking and Numerical Claim Verification	7
2.6	Multilinguality and Cross-Lingual Consistency	7
2.7	Numerical Reasoning in Language Models	8
2.8	Evaluation Metrics and Agreement Statistics	8
3	Related Work	9
3.1	LLMs as Rankers and Rerankers	9
3.2	Biases in LLM-as-a-Judge Evaluation	9
3.3	Automated and Multilingual Fact-Checking	10
3.4	Language Invariance and Cross-Lingual Consistency	10
3.5	Cross-Lingual Representations and the Latent Language	11
3.6	Numerical Reasoning Across Languages and Value-Aware Representations	11
3.7	Preference Optimisation and Parameter-Efficient Fine-Tuning	12
3.8	Positioning of This Thesis	12
4	Methodology	13
4.1	Task Formulation	13
4.2	Data	14
4.3	Trace-Scoring Architecture	14
4.4	Trace-Classification Objective	15
4.5	Language-Invariance Objective	16
4.6	Value-Aware Numerical Components	18
4.7	Baselines	19
4.8	Evaluation Protocol	20
4.9	Experimental Configuration	22
5	Results and Discussion	23
5.1	Experimental Scope and Data Hygiene	23
5.2	Task Performance	25
5.3	Cross-Lingual Consistency	26
5.4	Does Invariance Training Improve Ranking? (RQ1)	27
5.5	Where the Cross-Lingual Differences Live	28
5.6	Ablations	29
5.7	The Relationship Between Invariance and Reasoning Quality (RQ2)	31
5.8	Threats to Validity	32

5.9	Summary of Findings	32
6	Conclusion	33
6.1	Answers to the Research Questions	33
6.2	Contributions Revisited	34
6.3	Limitations	34
6.4	Future Work	35
6.5	Closing Remarks	35
	References	35
A	Full Per-Model Results	38
B	Prompt Templates	40
C	Technical Information	42
D	AI Usage Card	43

I hereby declare that I have written this thesis independently without any help from others and without the use of documents or aids other than those stated. I have mentioned all used sources and cited them correctly according to established academic citation rules.

Göttingen, June 10th, 2026

Last in Translation? Evaluating Language-Invariant Reason Ranking in Numerical Fact-Checking

AHMAD RAZA KHAWAJA, University of Göttingen, Germany

Numerical claims are common in political discourse, news, and social media, and verifying them is difficult even in one language. Multilingual verification is harder still. Large language models do not always treat translated versions of the same claim consistently. As a result, a claim and its translation may receive different reasoning-trace rankings and possibly different verdicts. This thesis asks whether language invariance is useful as a *training objective* for multilingual reasoning, rather than only as a property to measure.

We study the question on CLEF CheckThat! 2026 Task 2, which requires ranking candidate reasoning traces for numerical claims in English, Spanish, and Arabic. A verdict is then derived from the top-ranked traces. We build on our shared-task submission, a Llama-3.1-8B trace scorer fine-tuned with Quantized Low-Rank Adaptation (QLoRA), which ranked first on English and third on Spanish. We add a score-level language-invariance objective. It penalises the model when it gives different normalised relevance scores to corresponding reasoning traces across languages. We also introduce an evaluation protocol that separates task accuracy from cross-lingual agreement and measures agreement at two levels, the final verdict and the underlying trace ranking. We evaluate 28 configurations across three open-weight backbones on a held-out trilingual test set of 4,233 parallel claims.

The results show a clear separation between ranking quality and ranking stability. Invariance training does not change the official ranking metric, Recall@5 is statistically indistinguishable from standard multilingual fine-tuning, and both approaches are close to the metric’s attainable ceiling. However, invariance training makes the rankings roughly four times more stable across languages. The stability ranges of the invariance-trained and non-invariance systems do not overlap. Verdict quality also improves modestly, mainly in Arabic and especially for the minority Conflicting class. Verdict-level agreement is already high for every system and therefore does not distinguish between them. Measuring consistency only from final answers would have missed the main effect. The objective does not mainly change the label the model reaches, it changes whether the model relies on the same evidence across languages.

1 Introduction

1.1 Motivation

Numerical claims are pervasive in political discourse, news reporting, and social media. They typically involve quantities, dates, percentages, rankings, or comparisons. A small change in scope or numerical interpretation can completely change whether such a claim is true. Verifying numerical claims is therefore more demanding than standard textual entailment. A system must understand both the linguistic context and the quantitative relationships, rather than relying on surface meaning alone [28, 35]. The challenge grows in multilingual settings, where the same quantitative claim can appear in several languages and a reliable verifier must preserve its numerical meaning across them [15].

Large language models (LLMs) have become the standard tool for the reasoning and ranking components of automated fact-checking, used both as rankers of candidate evidence and as judges of reasoning quality [32, 37]. A growing body of work shows, however, that their judgments are not language-invariant, semantically equivalent inputs may receive different rankings or verdicts depending solely on the language in which they are expressed [12]. Prompt-based rankers also suffer from presentation biases. For example, the position or length of a candidate can affect the model’s choice [43]. At the same time, evidence suggests that multilingual LLMs learn partly language-independent internal representations [45, 54]. This creates a puzzle, why does shared internal structure not lead to consistent behaviour across languages?

This puzzle motivates the central question of the thesis. If a model is trained to make consistent judgments across languages, does its reasoning become more robust? Or does only its surface behaviour become more stable,

without any improvement in quality? Put differently, is language invariance a useful *objective* to optimise, or only a property worth measuring?

1.2 Problem Statement and Research Gap

An important point from the consistency literature makes this question more precise, consistency and correctness are separate properties [9, 30]. A model can give the same wrong answer in every language. It can also be accurate on average while giving different answers to translations of the same input. Cross-lingual agreement therefore says nothing about correctness on its own. Any benefit from invariance must be shown with an independent measure of task quality, rather than inferred from agreement statistics.

Two strands of recent work are relevant but have not yet been brought together. First, objectives that target cross-lingual consistency directly, consistency regularisation, cross-lingual self-alignment, and preference optimisation adapted to parallel prompts [25, 52, 56] show that invariance can be optimised rather than only observed. However, these methods are evaluated on factual question answering or general benchmarks, not on ranking quality in a downstream task. It therefore remains unclear whether optimising for invariance improves task performance. Second, standard subword tokenisation does not explicitly encode numerical magnitude, and LLMs often reason unreliably about quantities [34, 51]. The numeral *script* adds another source of cross-lingual variation. For example, the same value written with Arabic-Indic digits is tokenised differently and handled less reliably [36]. Value-aware numerical representations may reduce this problem [8], but it is not yet known whether they improve cross-lingual robustness as well as monolingual numeracy.

We therefore lack a systematic study of how a language-invariance objective interacts with improved numerical representations in multilingual numerical fact-checking. This thesis addresses that gap by making language invariance an explicit training target. It then measures accuracy and consistency separately to determine what this objective actually improves.

1.3 Grounding: CLEF CheckThat! 2026 Task 2

We study these questions in the concrete setting of CLEF CheckThat! 2026 Task 2 on multilingual numerical claim verification [48]. The task adopts a test-time scaling formulation for fact verification [4]. For each claim and its retrieved evidence, an LLM generates several reasoning traces at different temperatures to produce diverse reasoning paths. Redundant traces are removed. Participants then train a verifier to rank the remaining traces by how useful they are for reaching the correct verdict. A claim-level verdict is derived from the top-ranked traces. The English claims come from the QuanTemp benchmark [47], together with Spanish and Arabic claims from the previous CLEF edition. Ranking quality is measured with Recall@5 and MRR@5 over traces whose verdict matches the gold label, claim-verdict prediction with macro-averaged F1, and the per-language score is the mean of the two.

This setting suits the research question for three reasons. First, it is multilingual and evaluated separately for each language, so cross-lingual behaviour can be observed directly. Second, its explicit ranking of reasoning traces shows both *what* a system concludes and *which evidence* it relies on. This intermediate ranking is where the main cross-lingual effects appear. Third, the claims are grounded in quantities that should remain constant across languages and phrasings [24]. This makes it easier to attribute inconsistency to the model rather than to a real change in meaning during translation.

The thesis builds on our shared-task submission [22]. That system fine-tunes Llama-3.1-8B-Instruct [27] with Quantized Low-Rank Adaptation [7, 14] as a multilingual supervised trace scorer, scoring each claim–trace pair independently and ranking traces by predicted relevance. This approach is more controllable than generative ranking. It ranked first on English and third on Spanish on the official leaderboard, providing a competitive baseline and an established pipeline for this thesis.

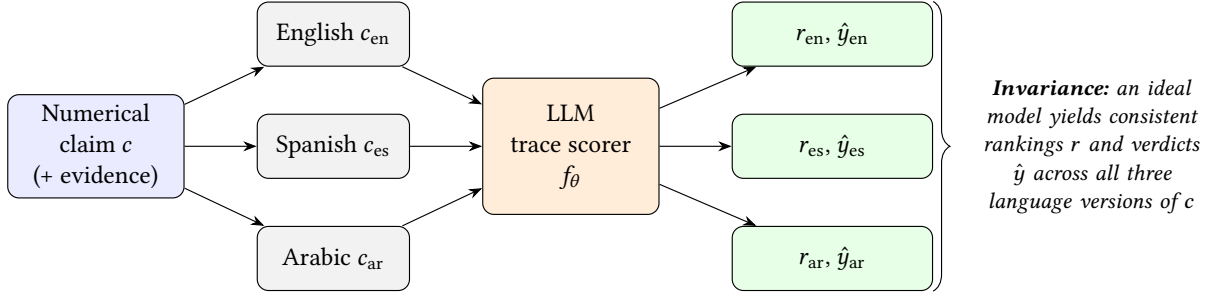


Fig. 1. The language-invariance problem in multilingual numerical fact-checking. The same numerical claim c and its evidence are presented to the trace scorer f_{θ} in three languages. A language-variant model may produce divergent rankings r and verdicts $\hat{y}$ for semantically equivalent inputs, whereas a language-invariant model should produce consistent outputs. This thesis asks whether explicitly optimising for such invariance improves ranking and reasoning quality.

Figure 1 illustrates the core problem. The same claim, evidence, and candidate reasoning traces are presented to the trace scorer in English, Spanish, and Arabic. A language-variant model may return different rankings and verdicts for these equivalent inputs. A language-invariant model should return consistent results. The empirical goal of this thesis is to measure and reduce these differences, and to test whether doing so also improves ranking quality.

1.4 Research Questions

Guided by the gap described above, this thesis investigates the relationship between language invariance and reasoning quality in multilingual numerical fact-checking through three research questions:

- RQ1** Does optimising for language-invariant reasoning improve ranking performance in multilingual numerical fact-checking?
- RQ2** What is the relationship between language invariance and reasoning quality across different languages?
- RQ3** Can language-invariant training objectives lead to more robust multilingual reasoning models?

The thesis also examines numerical representation. In particular, it tests whether value-aware representations, which make numerical magnitude more explicit than ordinary subword tokenisation, improve cross-lingual robustness. We treat this as part of RQ2 and RQ3 rather than as a separate research question because numerical grounding provides a direct test of language invariance.

1.5 Contributions

In addressing these questions, this thesis makes the following contributions:

- (1) It formulates language invariance as an explicit, differentiable *training objective*, rather than only an evaluation criterion. The objective operates on per-trace relevance scores and penalises differences between the normalised score profiles assigned to corresponding traces across languages. We apply it to a competitive trace-scoring pipeline derived from our CLEF CheckThat! 2026 Task 2 system.
- (2) It provides a systematic comparison of ranking formulations through the lens of cross-lingual consistency, spanning zero-shot listwise and pairwise prompted ranking, Direct Preference Optimization [33], and supervised trace scoring. It also explains why pointwise scoring makes invariance directly optimisable.
- (3) It contributes an evaluation protocol that reports task quality and cross-lingual agreement *independently*, and at two levels the final verdict and the underlying trace ranking and shows that these two levels give substantially different answers.

- (4) It reports an empirical study over three open-weight backbones and 28 non-degenerate configurations, evaluated on a held-out trilingual test set of 4,233 parallel claims with three seeds per configuration, including the interaction between the invariance objective and value-aware numerical representations.

Summary of findings. The main result is a separation between ranking quality and ranking stability. Optimising for language invariance leaves the official ranking metric essentially unchanged. Recall@5 is statistically indistinguishable from standard multilingual fine-tuning, and both approaches are close to the metric’s attainable ceiling. At the same time, the objective improves cross-lingual ranking stability roughly fourfold. The stability ranges of the invariance-trained and non-invariance systems do not overlap. Verdict quality also improves modestly, mainly in Arabic and especially in the minority Conflicting class. Two additional findings clarify this result. First, verdict-level agreement is high for every system and therefore cannot distinguish between them. An evaluation based only on final answers would have missed the effect. Second, a random baseline is perfectly invariant on every agreement measure but performs poorly on the task. Consistency must therefore be reported alongside task quality, never instead of it.

1.6 Thesis Outline

The remainder of this thesis is organised as follows. **Section 3** reviews related work on LLM-based ranking and its presentation biases, automated and multilingual fact-checking, language invariance and cross-lingual consistency, cross-lingual representations, numerical reasoning across languages, and the preference-optimisation and parameter-efficient fine-tuning methods on which the training objectives build. **Section 2** introduces the concepts and terminology the rest of the thesis assumes, for readers unfamiliar with language models, ranking formulations, fact-checking pipelines, or the agreement statistics used to quantify consistency. **Section 4** presents the methodology, the task formulation and data, the trace-scoring architecture and its training objective, the score-level language-invariance objective, the value-aware numerical components, the baselines, and the evaluation protocol. **Section 5** reports and discusses the results, presenting task performance and cross-lingual consistency separately before combining them to answer the research questions, and states the threats to validity. **Section 6** synthesises the findings, revisits the contributions, and outlines directions for future work. Full per-model results are provided in Appendix A.

2 Background

This section introduces the concepts and terminology used throughout the thesis. It assumes general machine-learning knowledge but not prior familiarity with LLMs, fact-checking, ranking, multilingual evaluation, or the metrics used here. We first explain language models and their adaptation to a task. We then introduce text ranking, binary classification, numerical fact-checking, cross-lingual consistency, and numerical reasoning. The section ends with the task and agreement metrics used in the experiments.

2.1 Large Language Models

A *large language model* (LLM) is a neural network that models the probability of natural-language text. Modern LLMs are usually decoder-only Transformers [49]. They learn by *next-token prediction*: given the preceding tokens, the model predicts the next one. For a sequence $x_1, \dots, x_T$, the joint probability is $\prod_{t=1}^T p_\theta(x_t \mid x_{<t})$. During *pre-training*, the model maximises this likelihood over a large text corpus. The resulting linguistic and world knowledge can then be adapted to specific tasks.

Tokens and tokenisation. LLMs process *tokens*, which are subword units produced by a *tokeniser*. The tokeniser is learned from the pre-training corpus. Common words may form one token, while rare words are split into several pieces. This scheme, often based on byte-pair encoding [41], keeps the vocabulary finite while representing any

string. Tokenisation matters here because numbers and non-Latin scripts may be split differently, which can affect numerical reasoning across languages (Section 2.7).

Instruction tuning. A raw pre-trained model predicts plausible continuations but may not follow instructions reliably. *Instruction tuning* fine-tunes it on instruction–response examples. The resulting models can follow natural-language requests. The three instruction-tuned *backbones* used here are Mistral-7B-Instruct [18], Llama-3.1-8B-Instruct [27], and Qwen3-8B [46]. They can rank items, classify text, or answer questions without task-specific training. The numbers 7B and 8B indicate billions of parameters, a rough measure of model capacity.

2.2 Adapting Language Models to a Task

An LLM can be adapted to a task in two broad ways. *Prompting* leaves its weights unchanged, whereas *fine-tuning* updates some or all of them. This thesis uses both approaches.

Prompting and in-context learning. In *zero-shot* prompting, the model receives only an instruction and the input. *Few-shot* prompting also provides several solved examples, allowing the model to infer the pattern through *in-context learning*. Prompting requires no gradient updates and is therefore a natural baseline. However, its output can be sensitive to wording and formatting.

Chain-of-thought reasoning. *Chain-of-thought* (CoT) prompting asks the model to produce intermediate reasoning steps before its final answer [53]. This often improves multi-step tasks such as arithmetic and multi-hop questions, while making the reasoning inspectable. The “reasoning traces” in this thesis are such chains of thought: each is a step-by-step argument over a claim and its evidence that ends in a proposed verdict. The task provides these traces. This thesis studies how to select and rank them, not how to generate them.

Full fine-tuning versus parameter-efficient fine-tuning. *Fine-tuning* continues training on labelled task data. Updating every parameter of a 7–8B model is expensive. *Parameter-efficient fine-tuning* (PEFT) instead freezes the pre-trained weights and trains a small set of new parameters. Low-Rank Adaptation (LoRA) [14] learns two small low-rank matrices for each adapted weight matrix. Their product is added to the frozen weight. Only these *adapters* are trained. The adapter rank r controls the number of trainable parameters, and α scales the update.

Quantisation and QLoRA. *Quantisation* stores weights at lower precision, such as 4 bits instead of 16 or 32, to reduce memory use. QLoRA [7] loads the frozen base model at 4-bit precision and trains higher-precision LoRA adapters on top. This makes billion-parameter fine-tuning feasible within the available GPU budget. Figure 2 illustrates the setup.

Preference optimisation. Models can also learn from *preferences*, a preferred and a rejected output for the same input. Reinforcement Learning from Human Feedback (RLHF) requires a separate reward model and reinforcement learning. Direct Preference Optimization (DPO) provides a simpler alternative [33]. It increases the likelihood of preferred outputs relative to rejected ones while keeping the model close to a reference model. This thesis uses DPO as a ranking baseline.

2.3 Text Ranking with Language Models

Ranking orders candidate items by how well they satisfy a criterion. *Re-ranking* applies this process to an already retrieved candidate set. LLM rankers use three common formulations, which differ in how many candidates the model sees at once (Figure 3).

- **Pointwise.** Each candidate is scored independently, and the ranking is obtained by sorting the scores. This is the formulation our trace scorer uses.

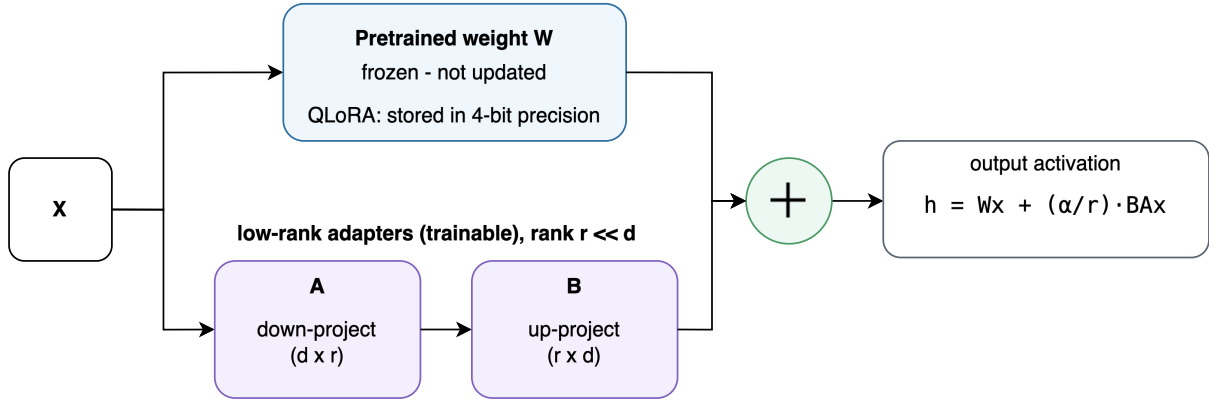


Fig. 2. LoRA and QLoRA. The pretrained weight matrix W is frozen a small pair of low-rank matrices A and B (with rank $r \ll d$) is trained and their product is added to the frozen weight, so the layer computes $h = Wx + (\alpha/r) BAx$ and only A and B are learned. QLoRA additionally stores the frozen weight W in 4-bit precision to reduce memory.

- **Pairwise.** The model compares two candidates at a time and states which is better, a global ordering is assembled from the accumulated pairwise preferences [32]. This keeps each decision simple but requires a number of comparisons that grows quadratically with the number of candidates.
- **Listwise.** All candidates are presented in a single prompt and the model emits a complete ordering. This is efficient at inference but couples the decision to the whole input, including the order in which candidates are listed [26].

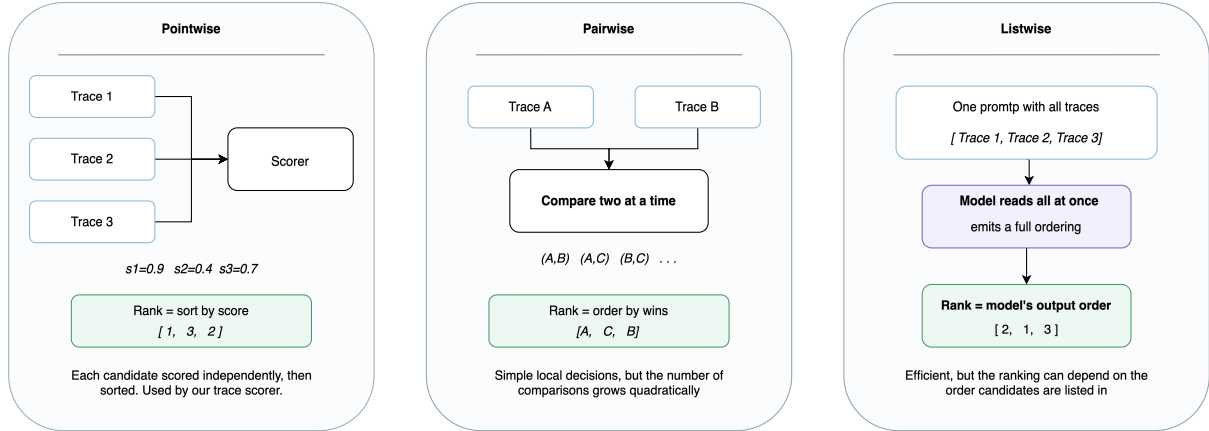


Fig. 3. The three standard formulations for ranking candidates with an LLM. *Pointwise* scoring evaluates each candidate independently and sorts by score (the formulation used by our trace scorer), *pairwise* ranking compares two candidates at a time and orders them by accumulated wins, *listwise* ranking presents all candidates in one prompt and reads off the model's output order.

LLM-as-a-judge and presentation bias. Using an LLM to evaluate candidates is often called *LLM-as-a-judge*. Such judgments can depend on factors unrelated to content quality. *Position bias* favours a candidate because of

where it appears in the prompt. Related effects include verbosity bias and sensitivity to prompt format [31, 43]. These biases motivate the move from generative ranking to independently scored candidates.

2.4 Classification and Training Objectives

Our system reformulates ranking as *binary classification*. A classifier maps an input to one or more real-valued *logits*. Binary classification can use one logit ℓ with the sigmoid function $\sigma(\ell) = 1/(1 + e^{-\ell})$, or two logits $(\ell^{\text{neg}}, \ell^{\text{pos}})$ with softmax. Both produce a probability for the positive class. Training minimises cross-entropy between this probability and the true label, the one-logit case uses binary cross-entropy. The positive-class probability becomes the relevance score used for ranking. A small *classification head* converts the model’s internal representation into the required logits.

2.5 Automated Fact-Checking and Numerical Claim Verification

Automated fact-checking assesses whether evidence supports a claim. A full pipeline usually includes *claim detection*, *evidence retrieval*, and *claim verification*. The first step identifies checkworthy claims, the second finds relevant documents, and the third assigns a veracity label from the evidence. This thesis assumes that claim detection and retrieval are complete and focuses on verification [37].

Core components: claims, evidence, and verdicts. A verification example contains a *claim*, retrieved *evidence*, and a *verdict*. The verdict describes the relationship between claim and evidence and is one of True, False, or Conflicting. The task adds a fourth component: candidate reasoning traces, each with its own proposed verdict, which the system must rank. A *numerical claim* depends on quantities such as counts, dates, percentages, rankings, or comparisons. Small changes in a value, unit, or comparison scope can reverse its verdict [47].

Reasoning traces and verifier-guided verification. *Test-time scaling* generates several independent reasoning traces instead of one answer. A separate *verifier* selects which traces to trust [4]. The task studied here follows this verifier-guided setup. It provides the claim, evidence, traces, and per-trace verdicts. The system ranks the traces and then derives a claim-level verdict from the highest-ranked ones. Because trace generation is outside the system, this is a ranking-and-aggregation task rather than free-form reasoning. Figure 4 shows the complete pipeline.

Benchmarks and the shared task. QuanTemp provides real-world numerical claims and evidence for open-domain verification [47]. CLEF CheckThat! 2026 Task 2 extends the verifier-guided formulation to English, Spanish, and Arabic [48]. Our system was developed in this setting [22]. Other multilingual benchmarks, including X-Fact [13] and XFEVER [3], likewise show that verification transfers poorly across languages and that performance varies widely.

2.6 Multilinguality and Cross-Lingual Consistency

A *multilingual* model is trained on several languages. The aim is *cross-lingual transfer*, an ability learned in one language should carry over to another. Multilingual models learn partly shared representations, so equivalent inputs can be encoded similarly. Interpretability studies of decoder-only LLMs suggest that non-English input passes through intermediate representations close to English before returning to the target language. This is described as a *latent language* [39, 54]. Such shared structure makes language-invariant behaviour possible in principle.

Consistency and language invariance. In practice, models are often inconsistent. *Consistency* means that meaning-preserving input changes produce non-contradictory outputs [9]. When the change is a translation, this becomes *cross-lingual consistency*, or *language invariance* [10, 30]. Consistency and accuracy are separate, a model can

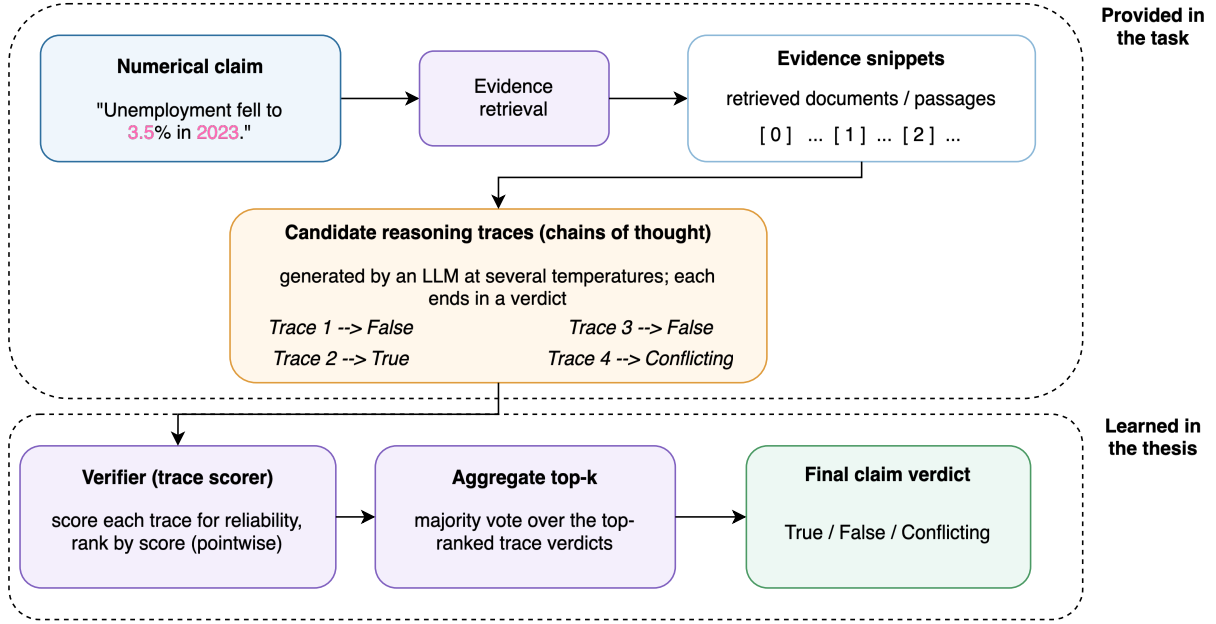


Fig. 4. Numerical claim verification framed as verifier-guided trace ranking. A numerical claim and its retrieved evidence, together with a set of candidate reasoning traces (each ending in a proposed verdict), are provided in the task. The system this thesis develops scores and ranks the traces, aggregates the top-ranked verdicts by majority vote, and outputs a final claim verdict from {True, False, Conflicting}.

be consistently wrong, or accurate on average while disagreeing across translations [9, 30]. Matching answers also do not prove that knowledge is represented identically inside the model [17]. The evaluation must therefore measure invariance and accuracy separately.

2.7 Numerical Reasoning in Language Models

Numeracy is the ability to reason about numbers, and it remains a weak point of language models. Models often fail on elementary operations [35, 51]. One reason is that standard token embeddings do not explicitly encode numerical magnitude. Subword tokenisation can also split a numeral into pieces that obscure its value [51]. *Value-aware* representations address this by encoding the numeric value of a span [8, 40]. Script and formatting create a separate problem. Arabic-Indic digits and different decimal or thousands separators lead to different tokenisations, and accuracy can fall even when the arithmetic is unchanged [1, 36]. This makes numeral script an important source of variation for Arabic.

2.8 Evaluation Metrics and Agreement Statistics

Finally, we define the metrics used to report results. They fall into three groups, classification quality, ranking quality, and cross-lingual agreement.

Classification metrics. For one class, *precision* is the fraction of predicted members that are correct, while *recall* is the fraction of true members that are retrieved. Their harmonic mean is $F1 = 2PR/(P + R)$. *Macro-F1* averages F1 across classes and therefore gives each class equal weight. We use it for the three-class claim verdict.

Ranking metrics. For a ranked list, *Recall@k* is the fraction of relevant items in the top k , while *Precision@k* is the fraction of the top k items that are relevant. *Mean Reciprocal Rank at k* (*MRR@k*) uses the reciprocal rank of the first relevant item, or zero if none appears in the top k , and averages it over queries. A trace is relevant here when its verdict matches the gold claim label. *Recall@5* is the primary ranking metric.

Inter-rater agreement. We treat each language-conditioned verdict as a “rater” of the same claim. *Raw agreement* is the fraction of claims with matching labels, but it includes agreement expected by chance. Chance-corrected statistics address this. Cohen’s κ compares two raters [5], Fleiss’ κ handles more than two [11], and Krippendorff’s α can also tolerate missing data [23]. Each equals 1 for perfect agreement and is around 0 at chance level.

Rank-agreement. Rankings require rank-correlation measures. Kendall’s τ compares the number of concordant and discordant item pairs [20], while Spearman’s ρ correlates the assigned ranks [44]. Both range from -1 for reversed order through 0 for unrelated order to $+1$ for identical order. Jaccard overlap compares the two top- k sets by dividing the size of their intersection by the size of their union. These measures show whether a model ranks the same traces similarly across languages, not only whether it reaches the same verdict. Section 4 uses these concepts to define the scorer, invariance objective, and evaluation protocol.

3 Related Work

This section reviews work on LLM-based ranking, multilingual fact-checking, cross-lingual consistency, numerical reasoning, and preference-based fine-tuning. Together, these areas frame the central question of the thesis, if a model is trained to behave consistently across languages, do its ranking and reasoning also become more reliable?

The review follows the three research questions. RQ1 concerns ranking performance, RQ2 concerns the relationship between invariance and reasoning quality across English, Spanish, and Arabic, and RQ3 concerns multilingual robustness. We first consider LLMs as rankers and judges, including their presentation biases. We then review multilingual fact-checking and cross-lingual consistency. The final parts cover multilingual representations, numerical reasoning, preference optimisation, and parameter-efficient fine-tuning.

3.1 LLMs as Rankers and Rerankers

LLMs are often used as rankers without task-specific training, especially in retrieval and evaluation. In pairwise ranking, the model compares two candidates at a time, and the comparisons are combined into a final ordering. Each decision is relatively simple, but the number of comparisons grows quadratically with the number of candidates [32].

Listwise ranking presents all candidates at once and asks the model for a complete ordering. It requires fewer inference calls, but its output depends on the entire prompt, including the order in which candidates appear. Recent work adapts learning-to-rank objectives to preference optimisation for more direct listwise training [26]. Even so, generative rankers remain sensitive to candidate order and prompt format, rather than only to candidate content [31].

This limitation motivates the independently scored design used in this thesis. The model scores each reasoning trace separately, and these scores determine the final ranking. The ranking can therefore reflect trace quality without depending on artefacts of a multi-candidate prompt.

3.2 Biases in LLM-as-a-Judge Evaluation

Similar problems occur when LLMs act as evaluators. Their judgments can depend on how options are presented, not only on content quality. Position bias makes a candidate more or less likely to be selected because of its location in the prompt. Verbosity bias can favour longer answers [43]. These biases can change the selected candidate even when its content is unchanged.

The main concern here is that LLM judges are often not language-invariant. The same evaluation can produce different judgments in different languages. A recent study of five model families, five tasks, and twenty-five languages found low multilingual judge agreement, with an average Fleiss’ κ of roughly 0.3. Neither multilingual pre-training nor larger model scale reliably removed the inconsistency [12]. This is the kind of cross-lingual instability studied in this thesis.

3.3 Automated and Multilingual Fact-Checking

Automated fact-checking is increasingly treated as an LLM reasoning and assessment problem. Recent work examines how task formulation affects the reliability of real-world claim verification, rather than asking only whether a model can produce a label [37]. Numerical claims are especially difficult because a small change in scope, units, dates, or interpretation can change the correct verdict.

QuanTemp pairs real-world, open-domain numerical claims with supporting evidence [47]. Later work extends this setting by generating several reasoning traces for each claim and using a verifier to select the most useful ones [4]. CLEF CheckThat! 2026 Task 2 adopts this verifier-guided setup for English, Spanish, and Arabic. Systems rank candidate traces and derive a claim-level verdict [48].

Broader multilingual benchmarks show the same difficulty. X-Fact contains naturally occurring, expert-labelled claims in twenty-five languages from eleven language families. Even strong models achieve only modest accuracy, showing that fact-checking ability transfers poorly across languages [13]. XFEVER translates FEVER claims and evidence into several languages to test zero-shot and translate-train verification. Performance varies substantially by language and generally remains below English [3]. X-FACTR reaches a similar conclusion for factual recall. Its native-speaker-authored queries in twenty-three languages reveal clear cross-lingual differences [19].

Together, these studies show why multilingual fact-checking is both important and difficult. They also provide the practical motivation for this thesis, if fact-checking systems behave differently across languages, then language invariance becomes a useful objective to study directly.

3.4 Language Invariance and Cross-Lingual Consistency

Language invariance requires a model to give consistent judgments when the same claim is expressed in different languages. This is central to a task that presents equivalent numerical claims in English, Spanish, and Arabic. To study it experimentally, the form of consistency must be defined and measured separately from accuracy.

Defining consistency. Consistency is commonly defined as stability under meaning-preserving changes to the input. A model is consistent if quasi-paraphrases of the same query produce non-contradictory predictions. The consistency score is the proportion of paraphrase pairs with agreeing predictions [9]. This definition separates consistency from accuracy. A model can be consistently wrong or accurate on average while giving inconsistent answers to equivalent inputs.

The same definition extends to translations. A language-invariant model should give the same output when two versions of a claim have the same meaning. However, mBERT and XLM-R are no more consistent across languages than a monolingual model is across English paraphrases. Consistency also decreases when typologically distant languages are included [10]. We therefore use *language invariance* to mean agreement across translated versions of the same input, measured independently of correctness.

Measuring cross-lingual consistency. Cross-lingual consistency therefore needs its own evaluation. Multilingual models often have low factual consistency across languages, and increasing model size does not reliably solve the problem. Consistency is also strongly related to subword-vocabulary overlap [30]. Script and tokenisation can therefore affect cross-lingual behaviour independently of reasoning ability.

The appropriate agreement measure depends on the output. For categorical verdicts, Cohen’s κ measures agreement between two raters [5], Fleiss’ κ extends it to more than two raters [11], and Krippendorff’s α can also handle missing data [23]. The multilingual LLM-as-a-judge work discussed above also uses Fleiss’ κ [12]. For rankings, Kendall’s τ and Spearman’s ρ measure rank correlation [20, 44], while Jaccard similarity measures overlap between top- k sets.

Behavioural agreement does not prove that a model uses the same internal representation across languages. Models can answer consistently while storing knowledge separately, especially across different scripts [17]. This thesis therefore treats invariance as an observable behavioural property, not as proof of a shared multilingual reasoning mechanism.

A measurement framing for this thesis. We measure language invariance at three levels. First, our primary evaluation is *verdict invariance*, which asks whether the predicted claim label remains stable across languages. For a claim c in $\mathcal{L} = \{\text{English, Spanish, Arabic}\}$, let $\hat{y}_\ell(c)$ be the verdict predicted from its language- ℓ version. We report the proportion of aligned claims for which all predictions agree. We also compute Fleiss’ κ over $\{\hat{y}_\ell(c)\}_{\ell \in \mathcal{L}}$, treating the three languages as raters of the same claim [11]. Pairwise Cohen’s κ [5] and Krippendorff’s α [23] provide supporting diagnostics.

As a secondary diagnostic, *ranking invariance* compares the trace orderings themselves. Let $r_\ell(c)$ denote the ordering produced for claim c in language ℓ . We compare these orderings with mean pairwise Kendall’s τ , Spearman’s ρ [20, 44], and top- k Jaccard overlap. These metrics show whether stability exists in the reasoning-trace ranking or only in the final verdict.

Third, training encourages *score-level invariance*. The model is penalised when it assigns different normalised score vectors to aligned traces across languages. This regulariser is related to the consistency objectives discussed in Section 3.7. Section 4 gives the exact formulation. All invariance measures remain separate from task accuracy. This is essential for RQ2 because a model may be consistent but wrong, or accurate but unstable [9, 30].

3.5 Cross-Lingual Representations and the Latent Language

Language invariance is plausible only if multilingual models share some representation of meaning. Evidence suggests that they do. mBERT transfers in zero-shot settings even across scripts, although typological similarity affects the result [29]. Shared multilingual structure can also emerge without a shared vocabulary [55]. XLM-R shows that large-scale multilingual pre-training improves transfer across many languages [6], while explicit alignment can bring contextual representations closer together [2]. A recent survey reviews these alignment methods and their evaluation [16].

Decoder-only LLMs show a more complex pattern. Logit-lens studies suggest that models such as Llama process multilingual input through intermediate representations that are often closest to English before producing the target language. This has been described as a *latent language* [54]. Other work also places semantic decisions in a representation space close to English and finds interventions most effective there [39]. Representational similarity analyses suggest that reasoning is only partly aligned across languages [45].

These findings suggest that invariance training could encourage more consistent use of a partly shared semantic space. It may therefore act as an inductive bias for multilingual robustness (RQ3). However, equal answers do not necessarily imply equal internal representations [17]. Behavioural invariance is evidence of stability, not proof of a language-independent reasoning mechanism.

3.6 Numerical Reasoning Across Languages and Value-Aware Representations

Numerical reasoning is a known weakness of LLMs. Models can fail even on simple operations, a pattern described as a fragile number sense [35]. Smaller diagnostic tests reach the same conclusion [34]. NumGLUE and broader

quantitative-reasoning evaluations show that numerical tasks test whether a model understands quantities rather than only their surrounding text [24, 28].

Standard token representations do not directly encode numerical magnitude. The model sees a number as a token sequence and must infer its value from the surface form. Some embeddings capture magnitude surprisingly well, but subword tokenisation still limits exact numeracy [51]. Value-aware methods instead encode the value of a numeric span [8], while related methods expose canonical numerical information directly [40].

The problem becomes more important across languages. MGSM and other multilingual chain-of-thought evaluations show that quantitative reasoning quality changes with the language even when the arithmetic remains the same [42]. This matters directly for numerical claims in English, Spanish, and Arabic.

Changing only the numeral script or format can substantially reduce accuracy for underrepresented scripts. Tokenisation partly explains the effect, while targeted prompting and explicit numeral mapping can reduce it [36]. Models also struggle with the compositional structure of numerals across languages unless the relevant operations are made symbolically explicit [1].

Numeral script is therefore another source of cross-lingual variation. Numerical fact-checking provides a demanding test of invariance because the same value can appear in different scripts. Value-aware representations and numerical normalisation may improve both numerical accuracy and cross-lingual consistency, especially for Arabic.

3.7 Preference Optimisation and Parameter-Efficient Fine-Tuning

The training setup builds on preference optimisation and parameter-efficient fine-tuning. Direct Preference Optimization (DPO) increases the likelihood of a preferred output relative to a rejected one while keeping the model close to a reference policy. It therefore learns from pairwise preferences without a separate reward model or reinforcement learning [33].

Parameter-efficient methods make these experiments feasible under a modest GPU budget. LoRA freezes the base model and trains only small low-rank update matrices [14]. QLoRA further reduces memory use by loading the base model at low precision and training adapters on top of it [7]. These methods support the open-weight instruction-tuned models used here, Mistral-7B [18], Qwen3-8B [46], and Llama-3.1-8B-Instruct [27].

Recent objectives apply similar ideas to cross-lingual consistency. Zheng et al. [56] use consistency regularisation to penalise prediction changes under meaning-preserving perturbations, including code-switching and machine translation. Cross-lingual self-alignment samples answers in several languages, selects the most self-consistent answer as the preferred target, and applies DPO [52]. Direct Consistency Optimization aligns completion likelihoods across parallel prompts in different languages without an explicit reward model [25].

These methods show that preference optimisation can target consistency as well as general output quality. Most are evaluated on factual question answering or broad benchmarks, however. They do not test whether optimising for cross-lingual consistency improves ranking in numerical fact-checking. RQ1 addresses this gap.

3.8 Positioning of This Thesis

The literature reveals a clear gap. LLM rankers and judges are sensitive to candidate order and prompt format [31, 43]. They can also produce different verdicts or rankings for equivalent inputs in different languages. Larger models and multilingual pre-training do not reliably solve this problem [10, 12, 30]. At the same time, multilingual models learn partly shared, often English-centred representations [6, 54]. Language-invariant behaviour may therefore be achievable, although it would not by itself prove a shared internal representation [17].

The numerical setting adds another difficulty. Standard token representations capture numerical meaning only partly, and numerical reasoning remains fragile [35, 51]. Performance also depends on numeral script and format [36]. The model must therefore rank traces for quantities expressed in different linguistic and script

forms. Recent work shows that preference optimisation and consistency regularisation can target cross-lingual consistency rather than only output quality [25, 52, 56]. What remains unknown is whether this improves multilingual reasoning in a trace-ranking fact-checking task.

This thesis studies that question in CLEF CheckThat! 2026 Task 2. Systems rank reasoning traces and produce a verdict for numerical claims in English, Spanish, and Arabic. The research questions separate ranking performance, the relationship between invariance and quality, and broader multilingual robustness.

RQ1: Ranking performance. Prior work shows that consistency can be optimised directly [25, 52, 56], but most of this work evaluates factual question answering or general benchmarks. RQ1 asks whether adding a language-invariance objective to the trace scorer improves the official ranking metrics, rather than only making outputs more similar across languages. This connects to work on ranking bias [31, 32, 43], if rankings are affected by irrelevant input variation, reducing language-driven variation may make them more reliable.

RQ2: Invariance and reasoning quality. Consistency is not the same as accuracy [9, 30]. A model can be stable across languages but wrong, or accurate in one language while unstable across others. RQ2 measures invariance and per-language reasoning quality separately and then examines their relationship across English, Spanish, and Arabic. Vocabulary overlap and script may affect consistency [30], and agreement need not imply shared internal representations [17]. The relationship between invariance and quality is therefore an empirical question.

RQ3: Multilingual robustness. The evidence reviewed above suggests that multilingual models share part of their semantic space [6, 39, 54]. RQ3 asks whether an invariance objective can use this structure as an inductive bias for robust multilingual reasoning. The experiments cover English, Spanish, and Arabic, with particular attention to the additional script and tokenisation variation in Arabic [36]. Value-aware representations and number-encoding methods provide complementary ways to reduce variation in the surface form of quantities [8, 40]. With this research gap established, Section 2 introduces the concepts and terminology needed to describe the proposed method precisely.

4 Methodology

This section describes how we test language invariance as a training objective for multilingual reasoning. We begin with the supervised trace-ranking system from our CLEF CheckThat! 2026 Task 2 submission [22, 48]. That system already contains two value-aware numerical components: numeric embeddings and plain-text numerical normalisation. Both aim to reduce variation in numeral script and format, especially for Arabic.

We extend the system in two ways. First, a score-level invariance objective encourages similar scores for corresponding traces across languages. This tests whether invariance improves ranking performance and robustness (RQ1 and RQ3). Second, the evaluation protocol measures invariance separately from task quality. Their relationship can therefore be observed rather than assumed (RQ2). We also test whether the inherited numerical components complement the invariance objective (RQ3).

The chapter proceeds from the task and data to the trace scorer and its classification loss. It then defines the invariance objective, numerical components, baselines, evaluation protocol, and experiment matrix for Mistral-7B, Llama-3.1-8B, and Qwen3-8B.

4.1 Task Formulation

We use the verifier-guided reranking formulation of CLEF CheckThat! 2026 Task 2 [48], which follows test-time scaling for numerical claim verification [4, 47]. Each example contains a claim c , retrieved evidence $E(c)$, and candidate reasoning traces $R(c) = \{1, \dots, n_c\}$. The traces were generated at several decoding temperatures to produce different reasoning paths. Each trace i has a verdict v_i , while the claim has a gold label y_c^* from $\mathcal{Y} = \{\text{True}, \text{False}, \text{Conflicting}\}$. The system outputs a trace ranking $r(c)$ and a claim verdict $\hat{y}_c \in \mathcal{Y}$.

The official evaluation is performed independently for each language $\ell \in \mathcal{L} = \{\text{English, Spanish, Arabic}\}$. Ranking quality is measured by Recall@5 over the traces whose verdict matches the gold claim label,

$$\text{Recall@5}(c) = \frac{|\{i \in r_{1:5}(c) : v_i = y_c^\star\}|}{|\{i \in R(c) : v_i = y_c^\star\}|}, \quad (1)$$

where $r_{1:5}(c)$ denotes the top five traces. Macro-F1 measures claim-verdict quality, and the official per-language score is the mean of macro-F1 and Recall@5. We also report MRR@5. Recall@5 has a structural ceiling. With m gold-matching traces, even a perfect ranking achieves at most $\min(5, m)/m$, when $m = 0$, Recall@5 is zero. On the validation data, the maximum attainable values are 0.39, 0.32, and 0.28 for English, Spanish, and Arabic. These ceilings limit the headroom of every ranker.

Generative rankers, listwise prompting, pairwise prompting, and DPO over complete rankings remain sensitive to candidate order and prompt format [12, 31–33, 43]. Supervised trace scoring instead assigns an independent relevance score to each claim–trace pair and sorts these scores deterministically. We use this scorer for the invariance objective because its continuous scores can be aligned directly across languages. A generative ranker would first require reconciling discrete orderings.

4.2 Data

We use the shared task’s multilingual numerical claim-verification dataset. The English portion is based on QuanTemp [47]. The Spanish and Arabic portions come from the previous edition and contain 2,808 and 3,260 claims, respectively, together with their evidence collections [48]. Each example includes a claim, evidence snippets, reasoning traces (Reasoning_traces), per-trace verdicts (Verdict_list), and a gold label (label). We map both verdict types to the common label space from Section 4.1. A trace is relevant when its normalised verdict matches the normalised gold label.

Complete and positive-only splits. Each training split has a *complete* and a *positive-only* version. The complete split keeps all claims. The positive-only split removes claims for which no trace has the correct verdict. Such a claim provides only negative examples to the binary scorer and increases class imbalance without teaching the distinction between useful and incorrect reasoning. Filtering is therefore applied only during training. Evaluation always uses the complete split, including claims for which Recall@5 must be zero.

Aligned multilingual data for invariance. The invariance objective requires parallel examples. We create translated versions of each claim, its evidence, and its traces. The original dataset_index and trace ordering are preserved, so trace i corresponds to trace i for the same claim in every language. This alignment allows their scores to be compared directly.

4.3 Trace-Scoring Architecture

Backbone models. The trace scorer uses three open-weight, instruction-tuned decoder models: Mistral-7B-Instruct-v0.3 [18], Llama-3.1-8B-Instruct [27], and Qwen3-8B [46]. Each backbone is loaded as a sequence-classification model. The pretrained transformer processes the claim–trace input, and a lightweight linear head maps its final representation to a relevance score.

The three families differ in their pre-training data and tokenisers. They therefore test whether the invariance effect generalises beyond one backbone, which is important for RQ2 and RQ3. Prior work links cross-lingual consistency to subword-vocabulary overlap [30, 36].

Scoring head. For a claim–trace input x_i , the model produces a relevance score s_i . The default cross-entropy head produces two logits, $(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})$. The corresponding probability that the trace is relevant is

$$p_i = \text{softmax}(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})_{\text{pos}}.$$

For ranking, we use the logit margin

$$s_i = \ell_i^{\text{pos}} - \ell_i^{\text{neg}},$$

which gives the same ordering as the positive-class probability. The alternative binary cross-entropy head produces one logit ℓ_i . It uses $s_i = \ell_i$ and $p_i = \sigma(\ell_i)$.

Both heads score each trace independently of the positions of other candidates. The continuous scores support both within-claim ranking and cross-lingual alignment.

Parameter-efficient fine-tuning. We fine-tune each backbone using QLoRA [7, 14]. The base model is loaded in 4-bit NF4 precision with a bf16 computation type, and its original parameters remain frozen. Trainable low-rank adapters are added to all linear projections using `target_modules=all-linear`. Unless stated otherwise, the adapters use rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, and no adapter bias.

The classification head remains outside the quantised model and is trained in full precision by setting `modules_to_save` to ["score"]. The new head must learn the relevance-score scale. PyTorch scaled dot-product attention and gradient checkpointing reduce memory use for long claim–evidence–trace inputs.

This configuration fits 7–8B models within the available multi-GPU budget. It also leaves most pre-trained multilingual representations unchanged, allowing us to test whether their shared latent structure can support invariant scoring [39, 54].

Input construction. Figures 5 and 6 show the full scoring process. Each claim–trace pair becomes one input:

```
Claim: {claim}
Evidence snippets:
[0] {evidence snippet}
[1] {evidence snippet}
...
Verdict: {trace verdict}
Justification: {cleaned trace justification}
```

We remove “Label:” and “Justification:” markers, repeated verdict tokens, and unnecessary whitespace from the justification. The verdict remains in its dedicated field. Removing duplicate label text prevents it from dominating the reasoning.

We limit the claim, each evidence snippet, the justification, and the number of evidence snippets. The final sequence is truncated to the configured token limit. The compact setting uses two evidence snippets and at most 1,024 tokens. The long-context setting uses up to twenty snippets, up to 3,078 characters per snippet, and at most 15,000 tokens. We exclude justifications shorter than three words because they contain too little reasoning for training.

4.4 Trace-Classification Objective

The trace scorer is a binary classifier. For trace i of claim c , the target is

$$z_i = \begin{cases} 1, & \text{if } v_i = y_c^*, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

A trace is relevant when its verdict matches the gold label. The two-logit head uses cross-entropy between $(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})$ and z_i . The single-logit head uses binary cross-entropy between $\sigma(\ell_i)$ and z_i . We call this term the ranking loss, $\mathcal{L}_{\text{rank}}$, because it trains the scores used for ranking.

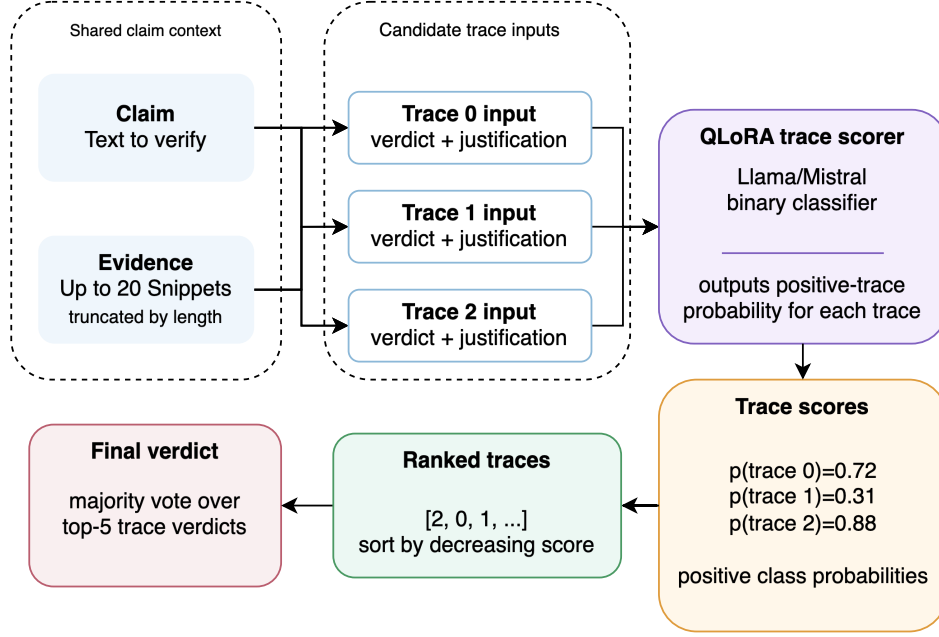


Fig. 5. Input-output flow of the supervised trace scorer. Candidate traces are scored independently, sorted by positive-class probability, and aggregated through the top-ranked trace verdicts.

At inference, traces are scored independently and sorted in decreasing score order:

$$r(c) = \text{argsort}_{i \in R(c)}(-s_i).$$

When two traces receive the same score, their original trace indices are used to break the tie deterministically.

The model does not use a separate head for the claim verdict. Instead, it takes a majority vote over the top- k trace verdicts, with $k = 5$ by default:

$$\hat{y}_c = \text{majority}(\{v_i \mid i \in r_{1:k}(c)\}). \quad (3)$$

If the vote is tied, the highest-ranked tied label wins. The learned trace scores therefore determine both the ranking and the claim verdict, so they can affect Recall@5 and macro-F1.

4.5 Language-Invariance Objective

The main methodological contribution is a score-level regulariser added to $\mathcal{L}_{\text{rank}}$. It penalises different relevance profiles for corresponding traces across languages. Unlike objectives defined only over discrete verdicts, it acts on continuous trace scores. It follows prior cross-lingual consistency work [25, 52, 56] but applies the idea directly to ranking, so its effect on ranking quality can be measured (RQ1). Figure 7 gives an overview.

Aligned multilingual batching. We build trace features separately for each language and tag them with a claim identifier (dataset_index), trace index, and language identifier. For every claim available in all languages, we keep the trace indices that survive preprocessing everywhere. These examples form an *aligned claim group*. Batches are made from complete groups, ensuring that the invariance loss has all required parallel scores. Claims with no common trace provide no invariance signal and are removed from the aligned set, although they may

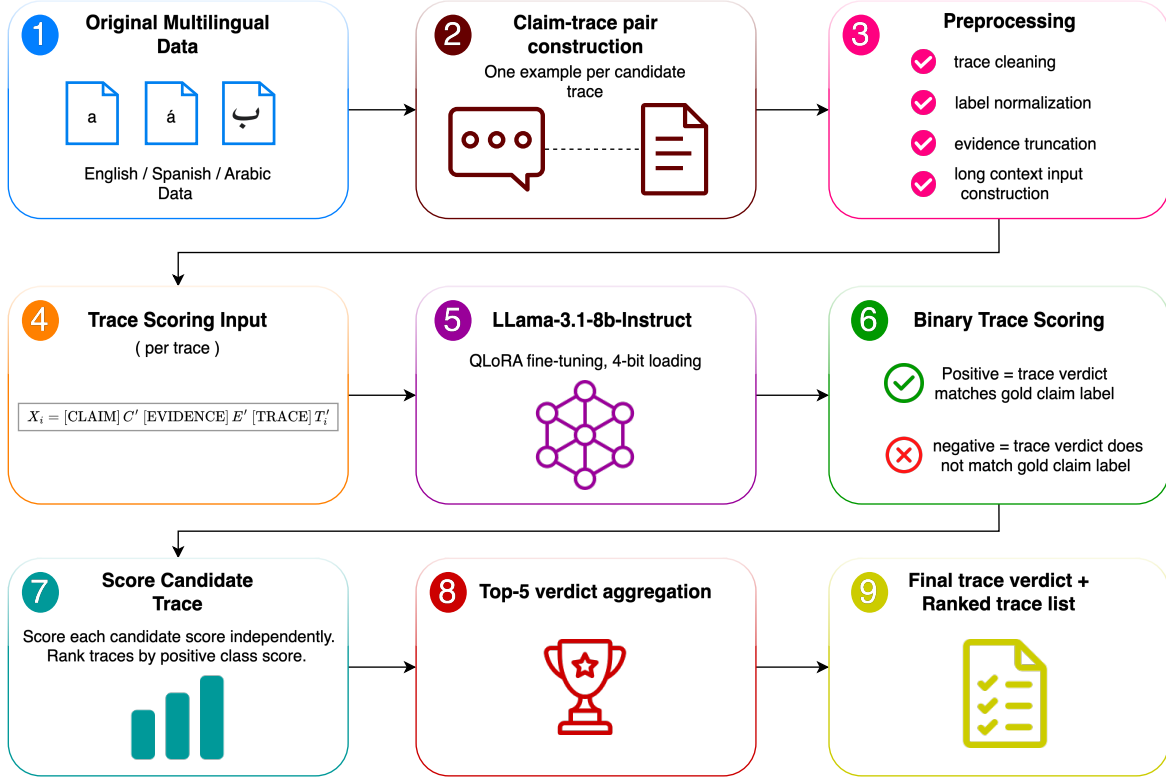


Fig. 6. Input-output flow of the supervised trace scorer. Candidate traces are scored independently, sorted by positive-class probability, and aggregated through the top-ranked trace verdicts.

still contribute to $\mathcal{L}_{\text{rank}}$. Classification labels and metadata remain unchanged, so the ranking loss matches the monolingual case.

Score normalisation. Absolute score scales may differ across languages even when the relative ordering is identical. Because ranking depends only on within-claim relative scores, we remove the per-claim, per-language location and scale. For claim c , language ℓ , and common trace set T_c , let $\mathbf{s}_c^{(\ell)} = (s_{c,i}^{(\ell)})_{i \in T_c}$. We standardise it as

$$\tilde{\mathbf{s}}_c^{(\ell)} = \frac{\mathbf{s}_c^{(\ell)} - \text{mean}(\mathbf{s}_c^{(\ell)})}{\text{std}(\mathbf{s}_c^{(\ell)}) + \varepsilon}, \quad (4)$$

where ε provides numerical stability. Standardisation removes harmless constant offsets and focuses on the shape of the score profile: which trace ranks above which and by how much relative to the spread. This aligns the regulariser with ranking behaviour.

Invariance loss. For each aligned claim with at least two languages, we compute the mean squared difference between the standardised score vectors for every language pair and average over claims:

$$\mathcal{L}_{\text{inv}} = \text{mean}_c \frac{1}{\binom{|\mathcal{L}_c|}{2}} \sum_{\ell < \ell'} \|\tilde{\mathbf{s}}_c^{(\ell)} - \tilde{\mathbf{s}}_c^{(\ell')}\|_2^2 / |T_c|, \quad (5)$$

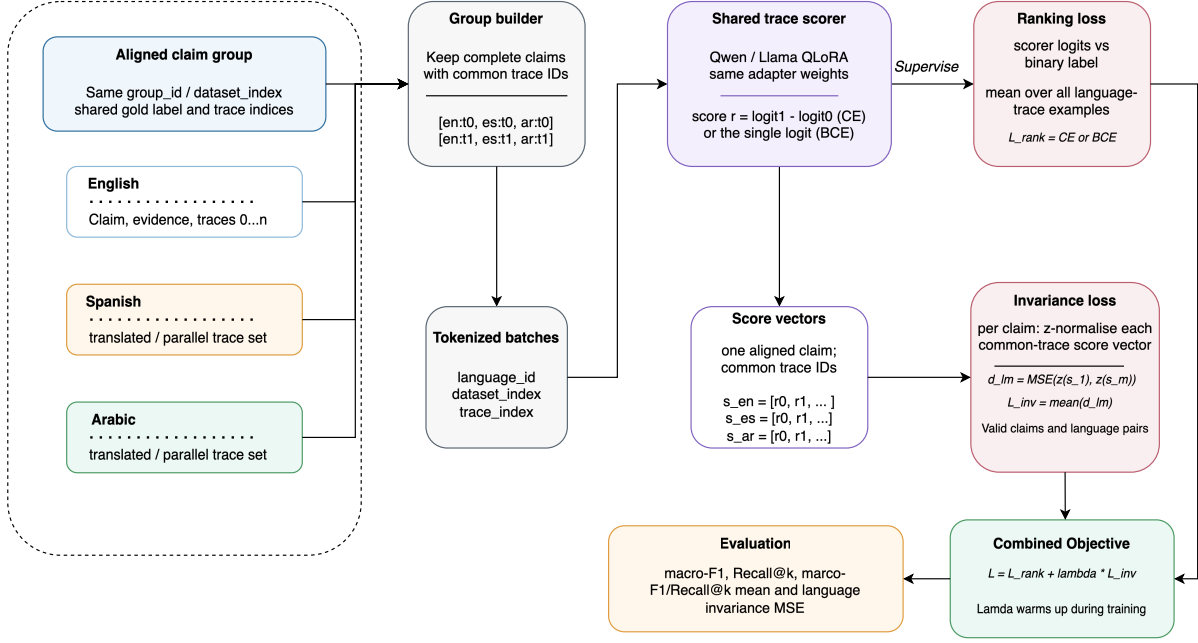


Fig. 7. Language-invariant multilingual trace-scoring pipeline. Parallel English, Spanish, and Arabic claim–trace examples are aligned by claim and trace indices and processed by a shared QLoRA scorer. Training combines binary trace-ranking loss with pairwise language-invariance loss computed over normalised, aligned score vectors.

where $\mathcal{L}_c$ is the set of languages available for claim c . The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_{\text{inv}}(t) \mathcal{L}_{\text{inv}}, \quad (6)$$

where the weight $\lambda_{\text{inv}}(t)$ increases linearly from 0 to its target during the first fraction ρ of the training steps and then remains constant:

$$\lambda_{\text{inv}}(t) = \begin{cases} \lambda^* \cdot \frac{t}{\rho T}, & t < \rho T, \\ \lambda^*, & t \geq \rho T, \end{cases} \quad (7)$$

where T is the total number of optimisation steps. Warm-up lets the head learn a useful score scale from $\mathcal{L}_{\text{rank}}$ before invariance pressure begins. Applying both losses from the first step could collapse the scores into a consistent but uninformative solution [9, 30]. Setting $\lambda^* = 0$ recovers the standard multilingual scorer, providing an ablation with the same data, batching, and architecture. We log $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{inv}}$, and $\lambda_{\text{inv}}(t)$ separately.

4.6 Value-Aware Numerical Components

The same quantity can have different surface forms across languages. It may use Western Arabic, Arabic-Indic, or Persian digits, as well as different decimal and thousands separators. These forms can produce different tokenisations and contribute to cross-lingual inconsistency [1, 36, 51]. To reduce this variation, we use two optional value-aware components. Both work with the standard scorer and the invariance objective.

Numeric canonicalisation. Both components use the same numeric canonicalisation. A regular expression identifies Western Arabic, Arabic-Indic, and Persian digits, percentage and per-mille signs, and numbers with

comma or decimal separators. Detected expressions are converted to ASCII digits. Their separators are interpreted and rewritten in one decimal format, while percentage and per-mille expressions are converted to decimal values. Each result is parsed and rendered as a canonical decimal string.

Different scripts and separator conventions can therefore map to the same value. The two components use this canonical form differently.

Value-aware numeric embeddings. Figure 8 shows the value-aware embedding pipeline. A special <num> token is inserted before each detected number, while the original number remains in the text. A small learned encoder represents the canonical value character by character using digits, a sign, a decimal point, and an exponent marker. A single-layer GRU processes these character embeddings and projects them to the backbone’s hidden size.

At each <num> position, the value vector replaces the standard token embedding. The transformer therefore receives both the explicit value and the original surface form. This reduces dependence on numeral tokenisation and follows previous work on value-aware numerical representations [8, 40].

The <num> token is added to the tokeniser, and the input embedding matrix is resized. The value encoder is trained with the LoRA adapters and classification head and saved with the trained adapter for inference.

Plain-text numeric normalisation. The second component exposes canonical values without changing the embedding layer. It appends a plain-text section headed `Normalized numbers`, to the trace-scorer input. This section lists the numbers found in the claim, evidence, and reasoning trace. When a number’s original and canonical forms differ, it is shown as a `surface=canonical` pair.

Because the normalised values are ordinary text, the architecture remains unchanged. This provides a lightweight alternative to the learned encoder and tests whether simply showing canonical values is enough. The procedure ignores small leading integers used as list markers, such as “1.” or “2)”, so they are not mistaken for quantities.

4.7 Baselines

The shared-task baselines provide reference points for both ranking quality and invariance.

- **Prompt baseline.** A frozen instruction-tuned backbone is the untrained reference. Given a claim, evidence, and candidate traces, it directly produces a ranking and verdict. We use both listwise and pairwise prompts.
- **Listwise prompting** (Figure 9). One prompt contains the claim, evidence, and every trace. The model returns strict JSON with a permutation of trace indices and one verdict. The ranking remains sensitive to the input order despite an instruction not to copy it, reproducing known position bias [31, 43].
- **Pairwise prompting** (Figure 10). The model compares two traces at a time, and accumulated wins define the ranking [32]. A *balanced* orientation alternates which trace appears first. A *bidirectional* orientation evaluates both orders and counts a win only when they agree. These methods reduce but do not remove order effects. They require $n(n-1)/2$ and $n(n-1)$ calls per claim, respectively.
- **Direct Preference Optimization (DPO)** (Figures 12 and 11). An English LoRA model starts from a supervised Mistral-7B adapter and is trained on automatically created preference pairs [33]. The chosen output ranks gold-matching traces first and predicts the gold verdict. Hard negatives either place a non-matching trace first, swap the top traces, or use an incorrect verdict. DPO operates on complete generated rankings rather than per-trace scores. It therefore retains order and format sensitivity and does not directly optimise Recall@5.

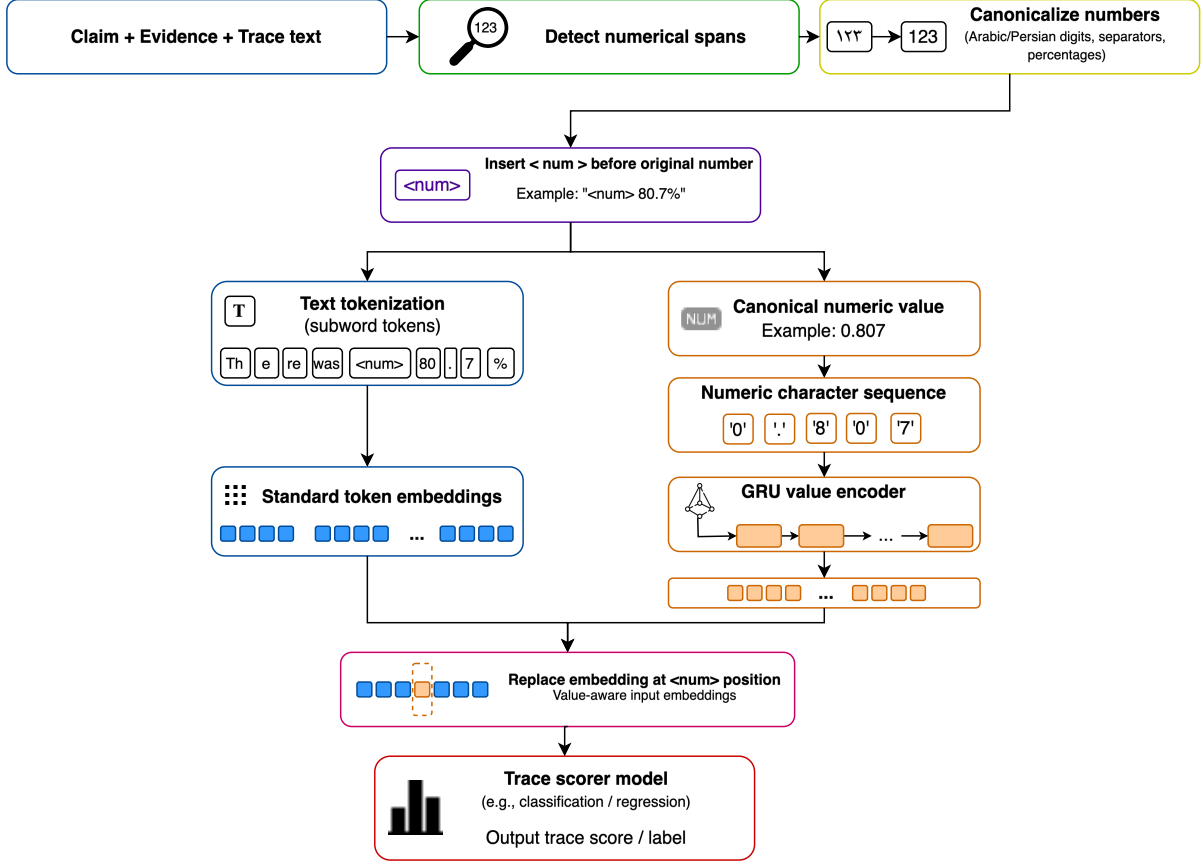


Fig. 8. Numeric embedding pipeline. Numerical spans are canonicalized, marked with `<num>`, encoded with a GRU value encoder, and inserted as value-aware embeddings before trace scoring.

4.8 Evaluation Protocol

We report task quality and language invariance *separately*. Their relationship is therefore an empirical result rather than part of the definition [9, 30] (RQ2).

Task metrics. For each language, we report Recall@5 and MRR@5 over gold-matching traces, verdict macro-F1, and the official mean of macro-F1 and Recall@5. We also report the Recall@5 upper bound. Multilingual runs give each metric per language and as a macro-average across languages.

Verdict invariance. We treat the language-conditioned verdicts $\{\hat{y}_\ell(c)\}_{\ell \in \mathcal{L}}$ as raters of the same claim. Agreement measures are the fraction of claims on which all languages agree, Fleiss’ κ across languages [11], pairwise Cohen’s κ [5], and Krippendorff’s α for missing variants [23]. Only aligned claims are compared.

Ranking invariance. Ranking agreement uses mean pairwise Kendall’s τ , Spearman’s ρ [20, 44], and top- k Jaccard overlap. The measures compare trace indices common to both orderings. The same evaluation also

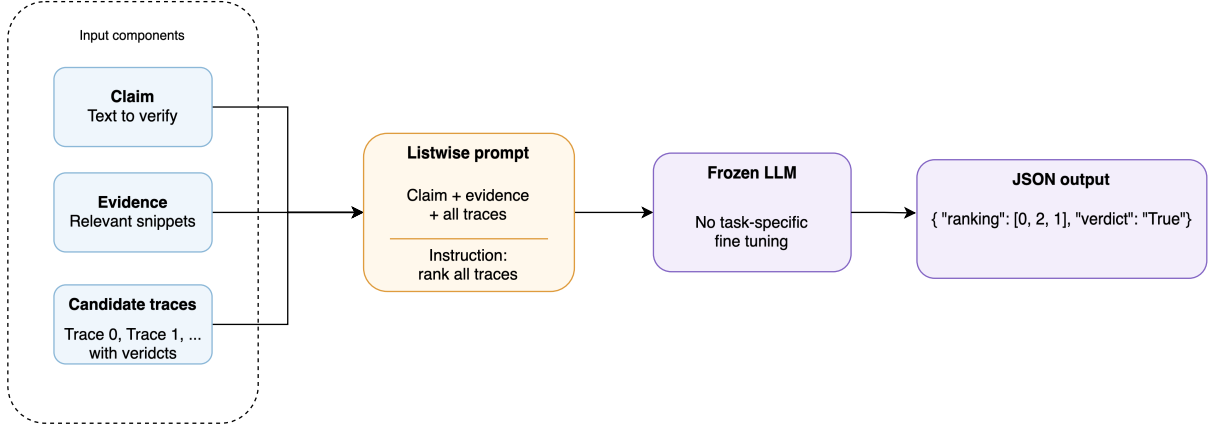


Fig. 9. Listwise prompted ranking setup. A frozen language model receives the claim, evidence, and all candidate traces in one prompt, then generates a complete ranking and final verdict.

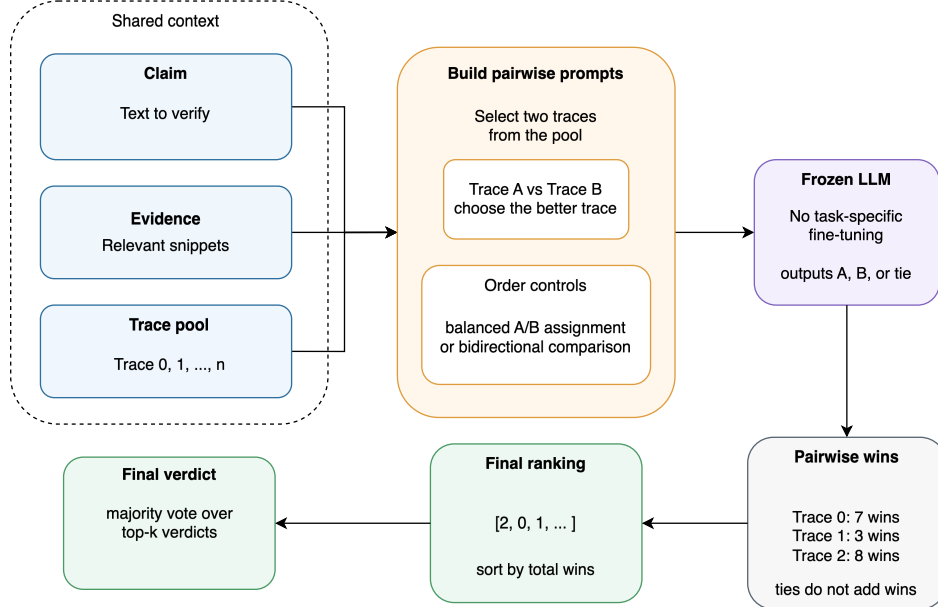


Fig. 10. Pairwise prompted ranking setup. A frozen language model compares two traces at a time, and accumulated pairwise wins are used to produce the final ranking.

compares original and translated variants, including validation sets with mismatched claim and trace languages. These checks help separate the effects of claim language, trace language, and translation noise.

Training-time invariance. On held-out data, the evaluator also reports $\mathcal{L}_{\text{inv}}$: the mean standardised score MSE from Equation (5), overall and by language pair. Comparing this training target with κ and τ shows whether lower score disagreement produces verdict- and ranking-level invariance without reducing task quality [17].

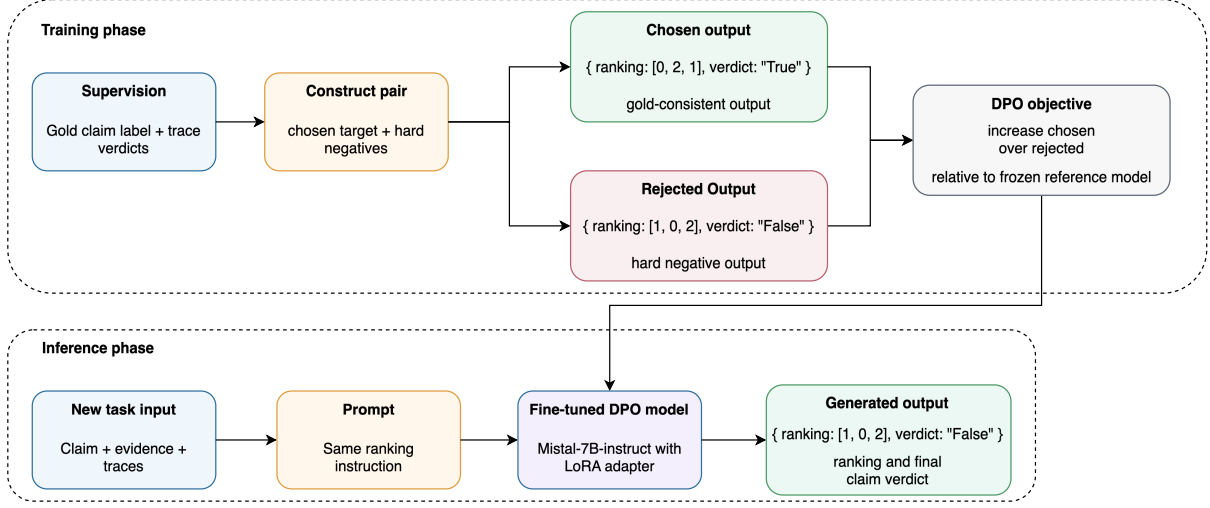


Fig. 11. DPO ranking setup. Chosen and rejected JSON outputs are automatically constructed from the gold label and trace verdicts, then used to train the model to prefer gold-consistent rankings.

4.9 Experimental Configuration

Training details. Unless stated otherwise, trace scorers train for five epochs with AdamW, a learning rate of 1×10^{-4} , a learning-rate warm-up ratio of 0.03, per-device batch size up to 8 with gradient accumulation, and bf16 compute. Early stopping uses positive-class F1 for single-language runs and the macro-F1/Recall@5 mean for multilingual runs. We save the adapter from the best evaluation step, not the final step. Invariance runs use $\rho = 0.2$ and sweep λ^* , always including $\lambda^* = 0$ as the control. The *regularised* setting uses adapter dropout 0.10 and weight decay 0.01. Every configuration uses three random seeds, and we report their mean. Weights and Biases records the experiments and metrics.

Compute. Experiments run on NVIDIA A100 GPUs. Main trace-scorer training uses two GPUs with distributed data parallelism through torchrun, inference and smaller ablations use one GPU. All LLM scorers load the base model at 4-bit precision, use QLoRA adapters and SDPA attention, and enable CUDA expandable memory segments for long inputs.

Experiment matrix. The backbones have different roles. **Mistral-7B-Instruct-v0.3** covers all shared-task ranking formulations and serves as a reference, it is not extended with the invariance objective. **Llama-3.1-8B-Instruct** is the primary backbone. It covers prompted baselines, DPO, multilingual trace scoring, long-context and regularised variants, numeric normalisation, and the invariance objective. **Qwen3-8B** uses the matching prompted baseline, the best Llama ranking setting (multilingual, long-context, and regularised), numeric normalisation, and invariance settings. This design separates the objective’s effect from one model family’s behaviour and compares English, Spanish, and the lower-resource, different-script Arabic setting (RQ2 and RQ3). The results section reports task quality and cross-lingual consistency separately before examining their relationship.

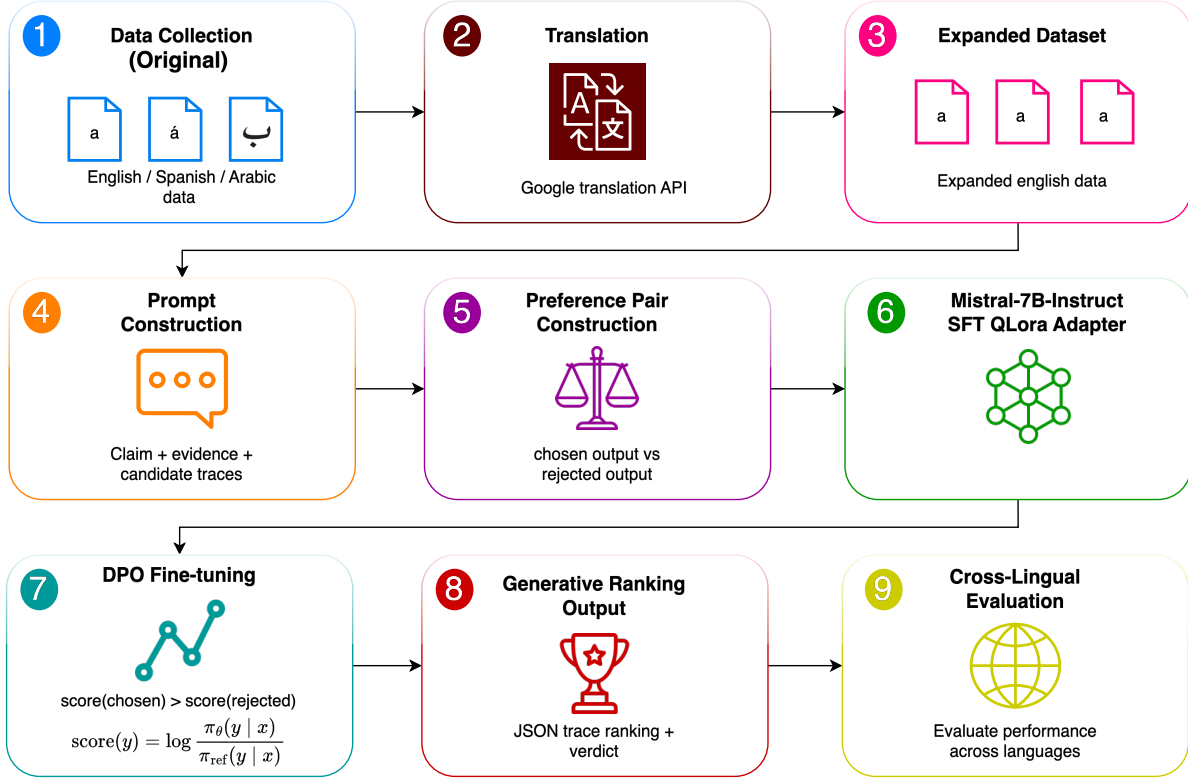


Fig. 12. DPO ranking pipeline. Translated multilingual data is converted into ranking prompts and chosen/rejected preference pairs for DPO fine-tuning, followed by cross-lingual evaluation.

5 Results and Discussion

This chapter reports and interprets the experimental results. We first describe the evaluation setup and explain which runs are included in the analysis (Section 5.1). We then report task performance (Section 5.2) and cross-lingual consistency (Section 5.3) separately. Section 5.4 brings these results together to answer RQ1. Section 5.5 identifies where the cross-lingual differences occur, and Section 5.6 examines the invariance weight, numeric normalisation, and backbone families. Section 5.7 analyses the relationship between invariance and reasoning quality (RQ2). Finally, Section 5.8 discusses the threats to validity, including a context-budget confound, and Section 5.9 summarises the findings for all three research questions.

5.1 Experimental Scope and Data Hygiene

All results use the held-out *test* set in the aligned trilingual form described in Section 4.2 and summarised in Table 1. It contains 4,233 claims, each available in English, Spanish, and Arabic with parallel trace sets of 20 candidate traces per claim. Task metrics are computed separately for each language, consistency metrics compare the three language-conditioned outputs for the same claim. The test set is separate from the training and validation splits. No configuration was trained on it, and early stopping and best-epoch selection used only the validation split. The results therefore measure generalisation to unseen claims. Each configuration was trained

and evaluated with three random seeds, and we report the mean. In total, the study contains 32 configurations, prompted baselines, earlier trace scorers, standard multilingual fine-tunes, and the invariance-trained models introduced in this thesis.

	Count	Share
Claims (each available in EN, ES, AR)	4,233	—
Candidate reasoning traces per claim	20	—
Claims with no gold-matching trace	578	13.7%
<i>Gold claim-label distribution</i>		
False	2,938	69.4%
True	868	20.5%
Conflicting	427	10.1%
Recall@5 upper bound	0.3005	

Table 1. Composition of the aligned trilingual test set. Every claim is present in all three languages with parallel trace sets, so the counts and the label distribution are identical per language. Claims whose candidate traces never match the gold label contribute zero Recall@5 whatever the ranking, which is what caps the attainable Recall@5 at 0.3005.

Excluded runs. Four of the 32 configurations are excluded from the statistics below. Three rows *Mistral, English DPO, Llama, inv. $\lambda^*=0.05$* , and *Qwen, prompted* report the same scores to four decimal places, identical across all three languages, with every cross-lingual agreement measure exactly 1.000. Inspecting their outputs explains the pattern, these runs degenerated to emitting the input order, returning the identity permutation $[0, 1, 2, \dots]$ for every claim instead of a content-based ranking. A ranking that ignores the input is by definition the same in every language, so on parallel data it yields identical per-language scores and perfect cross-lingual agreement. The resulting numbers describe the failure mode, not the systems, and the agreement of exactly 1.000 measures nothing at all. We therefore exclude these rows from every statistic and report them only for reference in Tables 2 and 3. The remaining analysis contains 28 configurations.

Direct Preference Optimization is a special case worth stating explicitly. It was attempted only in the single English configuration reported here. That run degenerated in the way just described and showed no advantage over the supervised trace scorer, so the approach was not developed further: no multilingual DPO variants, no additional seeds, and no hyper-parameter search were run. It appears in the tables as a record of the attempt rather than as a tuned baseline, and the comparison between preference-based and score-based ranking in this thesis should be read accordingly.

The fourth exclusion has a different reason and illustrates the central measurement principle of this thesis. The *random ranking* baseline also achieves perfect cross-lingual agreement Fleiss $\kappa = 1.000$ and Kendall $\tau = 1.000$ on every language pair. This is not an artefact, the baseline ranks traces in a fixed seeded permutation of trace indices that does not depend on the input at all, so it produces literally the same ranking in every language. It is therefore *invariant by construction*. However, it is also at the bottom of every accuracy measure. This shows that a system can be perfectly invariant while carrying no useful information. Invariance must therefore be reported alongside task accuracy, never instead of it [9, 30]. We exclude the random baseline from correlation statistics because its perfect agreement would dominate them without adding useful information. We retain it in Table 2 as the accuracy floor.

Achievable ceiling. As established in Section 4.1, Recall@5 has a structural upper bound. On this test set the bound is 0.3005 (Table 1). In addition, 578 of the 4,233 claims (13.7%) have no candidate trace whose verdict

matches the gold label. These claims receive zero Recall@5 regardless of the ranking. The Recall@5 results below should therefore be compared with 0.3005, not with 1.0.

Terminology. The task uses two metrics that behave differently, so we keep them separate. *Verdict quality* means claim-level macro-F1, while *ranking quality* means Recall@5. When one summary of task performance is needed, we use *mean macro-F1*, verdict macro-F1 averaged across English, Spanish, and Arabic. This is also what *accuracy* means below. It never refers to Recall@5 or the official combined score. *Consistency* refers only to cross-lingual agreement measures such as Fleiss κ , Cohen’s κ , and Kendall τ . These are calculated independently of both task metrics.

5.2 Task Performance

Table 2 condenses the results, for each training group it reports the two best configurations by mean macro-F1, placing accuracy and consistency side by side. Group averages over all configurations are quoted in the text where they support a statistical claim. Per-model results for all 28 configurations are given in Appendix A (Tables 3 and 4). Several observations follow.

Configuration	Rank	Recall@5 (ceiling 0.3005)				Macro-F1				Consistency	
		EN	ES	AR	Mean	EN	ES	AR	Mean	κ	τ
<i>Invariance-trained (this thesis)</i>											
Qwen, inv. $\lambda^*=0.1$	1	0.2663	0.2667	0.2622	0.2651	0.6192	0.6197	0.6157	0.6182	0.9479	0.5820
Llama, inv. $\lambda^*=0.2$	2	0.2699	0.2694	0.2667	0.2687	0.6211	0.6168	0.6161	0.6180	0.9471	0.5996
<i>Standard multilingual fine-tuning</i>											
Llama, standard	1	0.2649	0.2704	0.2692	0.2682	0.6256	0.6320	0.6105	0.6227	0.8810	0.0863
Llama, reg.	2	0.2710	0.2783	0.2673	0.2722	0.6263	0.6297	0.5808	0.6123	0.9095	0.1117
<i>Earlier trace scorers</i>											
Mistral, multiling. trace scorer (10 ep.)	1	0.2566	0.2549	0.2520	0.2545	0.5967	0.5932	0.5828	0.5909	0.9040	0.0955
Mistral, trace scorer	2	0.2196	0.2490	0.2171	0.2286	0.5847	0.5971	0.5775	0.5864	0.8404	0.0122
<i>Prompted baselines</i>											
Mistral, prompted (zero-shot)	1	0.2225	0.2173	0.2129	0.2176	0.5771	0.5795	0.5718	0.5761	0.9168	0.3674
Llama, prompted (zero-shot)	2	0.2234	0.2222	0.2219	0.2225	0.5751	0.5716	0.5792	0.5753	0.9049	0.0697
<i>Reported for reference only, excluded from every statistic (see Section 5.1)</i>											
Random ranking (floor)	—	0.2181	0.2181	0.2181	0.2181	0.5820	0.5820	0.5820	0.5820	—	—
Llama, inv. $\lambda^*=0.05^\dagger$	—	0.2228	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938	—	—
Mistral, English DPO [†]	—	0.2228	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938	—	—
Qwen, prompted (zero-shot) [†]	—	0.2228	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938	—	—

Table 2. Condensed results on the held-out trilingual test set, showing the two best configurations in each training group ranked by mean macro-F1. Bold marks the best value per column among the 28 valid configurations. Consistency is Fleiss κ over the language-conditioned verdicts and mean pairwise Kendall τ (EN–AR); both are dashed for the reference rows, whose perfect agreement is an artefact of ranking independently of the input (Section 5.1). Group means over all configurations are reported in the text; per-model results are in Tables 3 and 4.

Every trained system beats the baselines, but the margins are narrow. The random ranking baseline reaches 0.2181 Recall@5 and 0.5820 macro-F1 in English. The zero-shot prompted baselines are barely different, the Llama prompted baseline scores 0.2234 and 0.5751. This confirms, on a larger and fully parallel evaluation set, the shared-task finding that zero-shot generative ranking adds little over chance in this setting. Fine-tuned trace

scorers improve substantially on macro-F1, reaching up to 0.6263 in English. Their Recall@5 improvement is smaller, reaching up to 0.2710.

Ranking headroom is nearly exhausted, verdict quality is not. The best Recall@5, 0.2710, is roughly 90% of the achievable ceiling of 0.3005. The ranking component is therefore close to saturated for this candidate-trace set. Further gains will require candidate sets with more gold-matching traces, not only better ordering of the current traces. Macro-F1, by contrast, ranges from 0.5715 to 0.6320, so modelling choices still matter for verdict quality. The official score averages these two metrics and is therefore increasingly driven by the verdict term.

Metric redundancy. Across the 28 configurations, Recall@5, Precision@5, MRR@5, macro-F1, and the per-class F1 scores for True and False are almost collinear. Their Pearson correlations range from 0.92 to 0.99. F1 for the minority Conflicting class is the exception, with correlations of only $r = 0.61$ –0.85. Most headline metrics therefore produce nearly the same ordering of systems. The Conflicting class is what separates them. Section 5.5 shows that this class also carries the cross-lingual effect.

5.3 Cross-Lingual Consistency

The right-hand block of Table 2 reports consistency independently of accuracy, Table 4 gives the per-model values. The two types of consistency measure behave very differently. Only ranking agreement clearly distinguishes between systems.

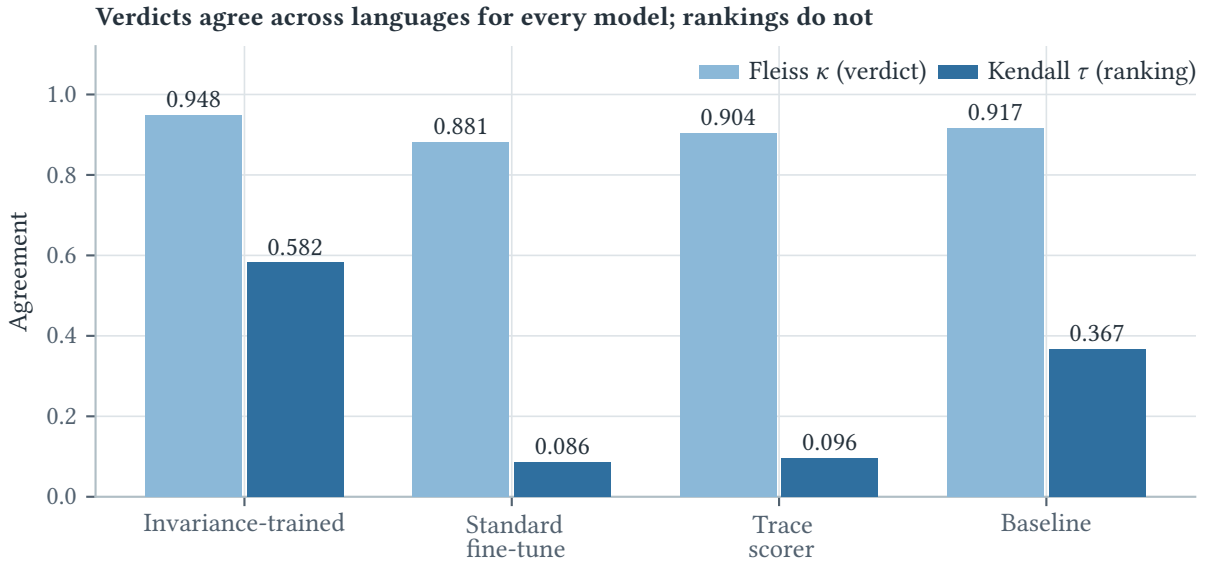


Fig. 13. Verdict agreement versus ranking agreement for the best configuration in each training group, selected by mean macro-F1. Fleiss κ over the three language-conditioned verdicts is high for every group and separates them weakly. Mean pairwise Kendall τ between the language-conditioned trace rankings separates the same systems by almost an order of magnitude.

Verdicts agree across languages, rankings do not. Fleiss κ over the language-conditioned verdicts is high and falls within a narrow range, on average 0.8596 for the earlier trace scorers, 0.9028 for standard fine-tuning, and

0.9443 for invariance-trained models. Figure 13 shows the same pattern for the best configuration in each group. Based only on verdict agreement, every system appears strongly language-invariant. Kendall τ between the trace *rankings* gives a very different picture. For English–Arabic, it ranges from 0.012 to 0.644 across individual configurations and from 0.0578 to 0.5667 across groups. For English–Spanish, the weakest configuration is slightly negative (-0.008), meaning that its ordering is no better than unrelated.

Standard multilingual fine-tuning therefore produces models that usually reach the same verdict in all three languages but support it with different evidence. The top-ranked traces in English and Arabic are nearly uncorrelated ($\tau \approx 0.13$). Because the final verdict is a majority vote over the top five traces, stable labels can result from unstable and largely different evidence. Verdict agreement therefore overstates cross-lingual robustness. Ranking agreement reveals the difference. If we had reported only κ over final answers, we would have concluded that all systems were already language-invariant.

5.4 Does Invariance Training Improve Ranking? (RQ1)

Figure 14 plots every configuration by mean macro-F1 (accuracy) and Kendall τ between English and Arabic rankings (stability). The two training groups separate clearly on stability but overlap on accuracy.

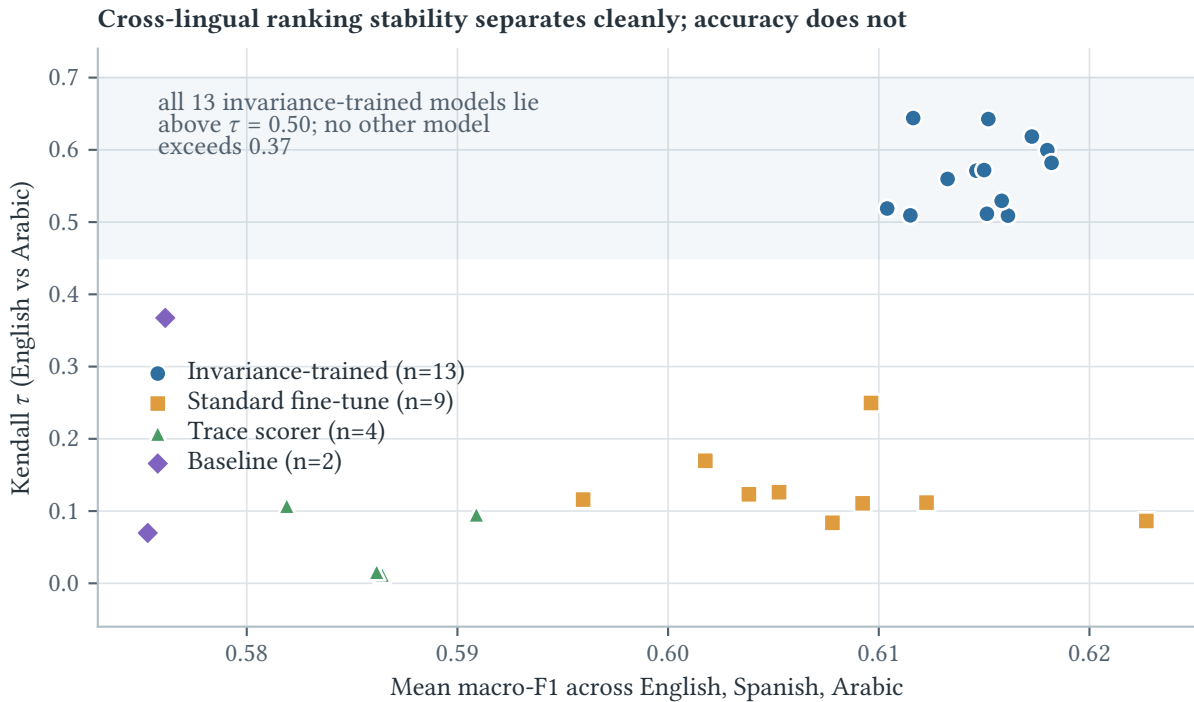


Fig. 14. Each point is one configuration ($n = 28$). The vertical axis is the mean pairwise Kendall τ between the English and Arabic trace rankings the horizontal axis is macro-F1 averaged over the three languages. The invariance-trained population is fully separated from every other configuration on stability, while the accuracy ranges overlap.

The stability effect is large and fully separated. Table 2 summarises the group-level picture. The 13 invariance-trained configurations reach a mean English–Arabic Kendall τ of 0.5667. The corresponding value is 0.1308 for

standard fine-tuning, 0.0578 for the earlier trace scorers, and 0.2185 for the baselines. The stability ranges do not overlap, the *worst* invariance-trained configuration ($\tau = 0.5089$) is still well above the *best* non-invariance configuration ($\tau = 0.3674$). The effect size is Cohen’s $d = 6.09$ (Welch $t = 16.71$, $p < 10^{-13}$, Mann–Whitney $p < 10^{-5}$). Verdict agreement moves in the same direction, from Fleiss $\kappa = 0.8924$ to 0.9443 ($d = 2.20$, $p < 10^{-4}$). This complete separation of the stability ranges is the main evidence for the effect.

The accuracy effect is positive but not the argument. Mean macro-F1 is also higher for invariance-trained models, 0.6148 compared with 0.5977 ($d = 1.61$, $p < 0.001$). Unlike the stability ranges, however, the accuracy ranges overlap. The best non-invariance configuration (*Llama, standard*, 0.6227) outperforms the best invariance-trained configuration (*Qwen, inv. $\lambda^* = 0.1$* , 0.6182). On English macro-F1 alone, invariance training shows no improvement, the best English score belongs to *Llama, reg.* (0.6263).

On the official ranking metric, the two are tied. Recall@5 needs to be considered separately because it is the official ranking metric and does *not* follow macro-F1. Averaged over the three languages, standard fine-tuning has a marginally *higher* Recall@5 than invariance training (0.2679 against 0.2656), and the single best Recall@5 of the study belongs to a standard fine-tune (*Llama, reg.*, 0.2710 in English). Per language the two groups are within 0.007 of each other everywhere (Figure 15, right panel). Given the ceiling of 0.3005 and the near-collinearity of the ranking metrics noted in Section 5.2, the fair reading is that the two approaches are tied on Recall@5. Invariance training does not measurably change how much relevant evidence appears in the top five.

The answer to RQ1 therefore depends on what is meant by improvement. Optimising for language invariance does *not* increase per-language Recall@5. On that metric, it matches standard fine-tuning and is slightly lower on average. It does, however, make the trace ordering stable under translation, improving cross-lingual ranking agreement roughly fourfold. It also improves verdict quality and reduces the loss in Arabic. The contribution is therefore greater robustness across languages, not higher state-of-the-art ranking performance.

5.5 Where the Cross-Lingual Differences Live

Across the 28 configurations, Spanish macro-F1 (0.6118) is slightly higher than English (0.6099). Arabic macro-F1 is 0.5951, which is 0.0147 below English, or about 2.4% lower. Recall@5 follows the same pattern for English, Spanish, and Arabic (0.2579 / 0.2620 / 0.2548). The multilingual difficulty in this task is therefore mainly an Arabic effect. This is consistent with the expectation from Section 2.7: a different numeral script and weaker representation in pre-training make Arabic the more difficult case [30, 36].

Figure 16 shows that the Arabic drop is not spread evenly across labels. Between English and Arabic, F1 falls by only 0.0026 for False and 0.0027 for True. For the minority Conflicting class, it falls from 0.2625 in English and 0.2671 in Spanish to 0.2236 in Arabic. This drop is roughly fourteen times larger than for either majority class. The difference becomes clearer when the models are grouped by training type. Standard fine-tunes fall from 0.2701 in English to 0.1781 in Arabic on Conflicting F1, a relative loss of 34.1%. Invariance-trained models fall only from 0.2698 to 0.2513, a relative loss of 6.8%. Unlike the aggregate Arabic gap discussed below, this difference is statistically clear (Welch $p = 0.0001$, Mann–Whitney $p = 0.0003$). Only two of the nine standard fine-tunes fall within the range of the thirteen invariance-trained models. Conflicting is the rarest class, with 427 of 4,233 claims (10.1%). It also requires the model to weigh contradictory quantitative evidence rather than match a single number. Unstable evidence selection is therefore especially harmful for this class. The class-level analysis explains the aggregate result, invariance training helps Arabic mainly by protecting the hardest and rarest decision.

A caution on the aggregate Arabic gap. The mean English-to-Arabic macro-F1 drop is -0.0098 for invariance-trained models and -0.0311 for standard fine-tuning. The difference is roughly threefold in the expected direction. However, this comparison does *not* reach statistical significance (Welch $p = 0.089$, Mann–Whitney $p = 0.231$),

Arabic is the weakest language; the invariance gain is in macro-F1, not Recall@5

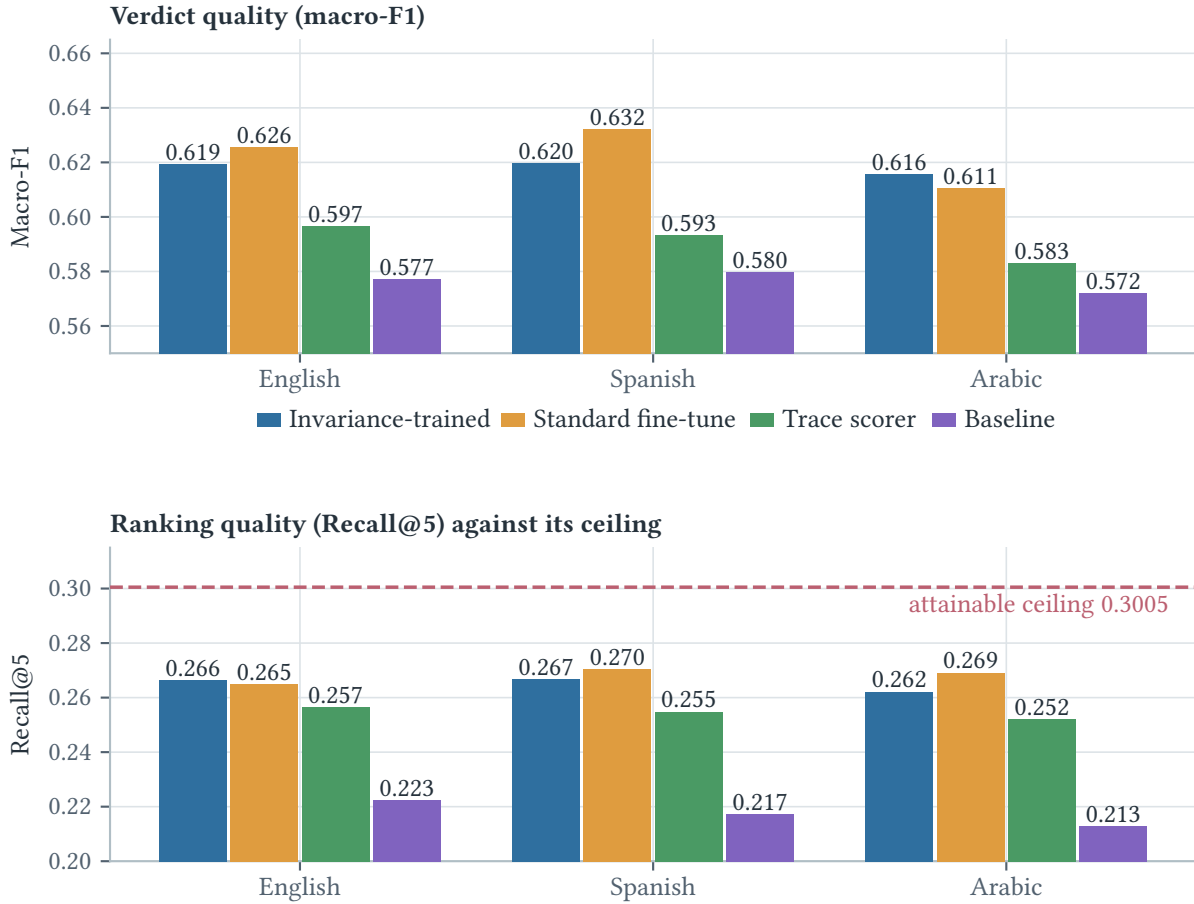


Fig. 15. Both official metrics by language, for the best configuration in each training group (selected by mean macro-F1). *Top*: macro-F1. Spanish is on par with English, Arabic is consistently the weakest, and the Arabic gap is smallest for the invariance-trained system. *Bottom*: Recall@5 against its attainable ceiling of 0.3005 (dashed). The two fine-tuned systems are level on Recall@5 and sit close to the ceiling, so the invariance effect visible in the upper panel does not appear in the lower one.

because the per-configuration variance in the gap is large relative to the sample size. The pattern is consistent across model families and settings, and the class-level analysis supports it. Nevertheless, it should be described as a trend rather than an established effect. The ranking-stability result, by contrast, is unambiguous.

5.6 Ablations

The invariance weight. Figure 17 sweeps the target invariance weight λ^* over $\{0.1, 0.2, 0.3\}$ for both backbones. Cross-lingual stability increases monotonically with λ^* for both model families. It rises from 0.5711 to 0.6440 for Llama and from 0.5820 to 0.6426 for Qwen. This is the expected result of increasing the regulariser strength.

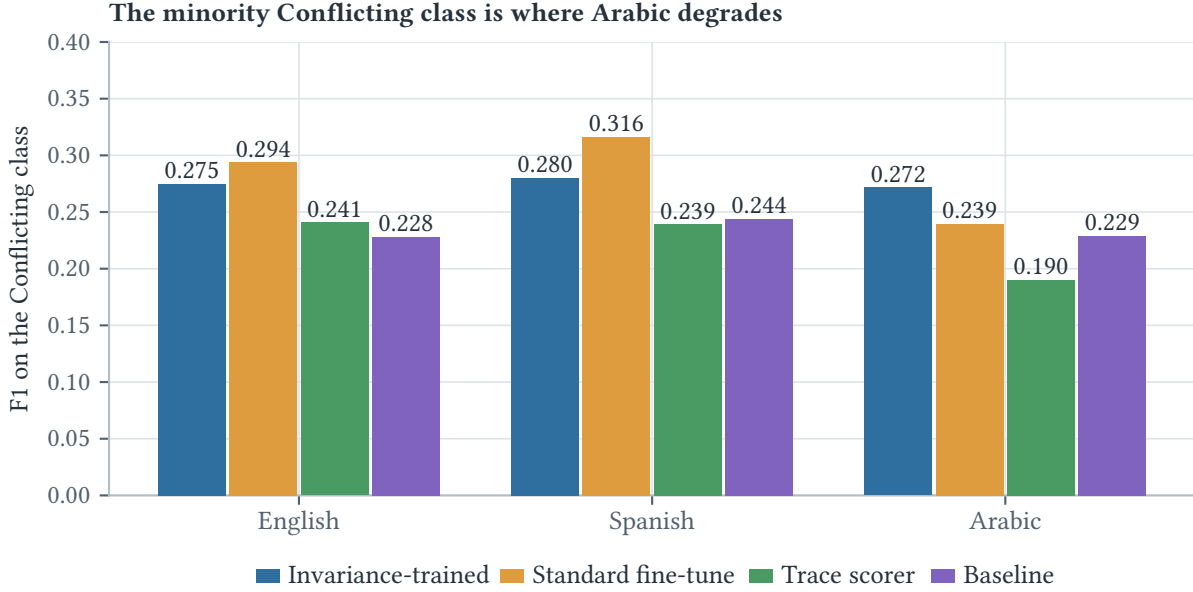


Fig. 16. F1 on the minority Conflicting class by language, for the best configuration in each training group (selected by mean macro-F1). The Arabic degradation seen in Figure 15 is concentrated almost entirely in this class, the standard fine-tune loses far more of its Conflicting F1 between English and Arabic than the invariance-trained system does. Group averages are reported in the text.

Mean macro-F1 does not increase in the same way. For Llama, it peaks at $\lambda^* = 0.2$ (0.6180, compared with 0.6146 at 0.1 and 0.6116 at 0.3). For Qwen, it declines slightly from 0.6182 to 0.6173 and then 0.6152. Recall@5 follows macro-F1 rather than τ and peaks at $\lambda^* = 0.2$ for both backbones. The sweep therefore shows a mild trade-off between task quality and consistency. The value $\lambda^* = 0.2$ provides the best balance. All tested values still produce much greater stability than the non-invariance systems, so using the objective matters more than the exact weight chosen within this range.

Numeric normalisation does not combine with invariance training. Plain-text numeric normalisation was carried over from the shared-task system as a complementary method for the numerical dimension of the task (Section 4.6). Within the invariance-trained group it does not help. Configurations combining the two reach $\tau = 0.5121$ and mean macro-F1 0.6133, against $\tau = 0.5910$ and 0.6154 for invariance training alone, adding numeric normalisation reduces Kendall τ by 0.079 and does not improve accuracy. The same pattern appears in Arabic (0.6028 compared with 0.6111). Outside the invariance group, the result is different, numeric normalisation gives a small improvement to standard fine-tunes in English and Spanish. The two methods therefore seem partly redundant rather than individually ineffective. Both address the fact that the same quantity can be represented differently across scripts. The invariance objective already forces scores for these forms to agree, so adding canonical values to the input may increase its length without providing new information. The evidence does not support combining the two methods.

Backbone families. Across all settings, Qwen3-8B and Llama-3.1-8B have similar accuracy, their mean macro-F1 scores are 0.6128 and 0.6083, respectively. Qwen is more stable across languages, with $\tau = 0.4505$ compared with

Stability rises monotonically with λ^* ; both task metrics peak at $\lambda^* = 0.2$

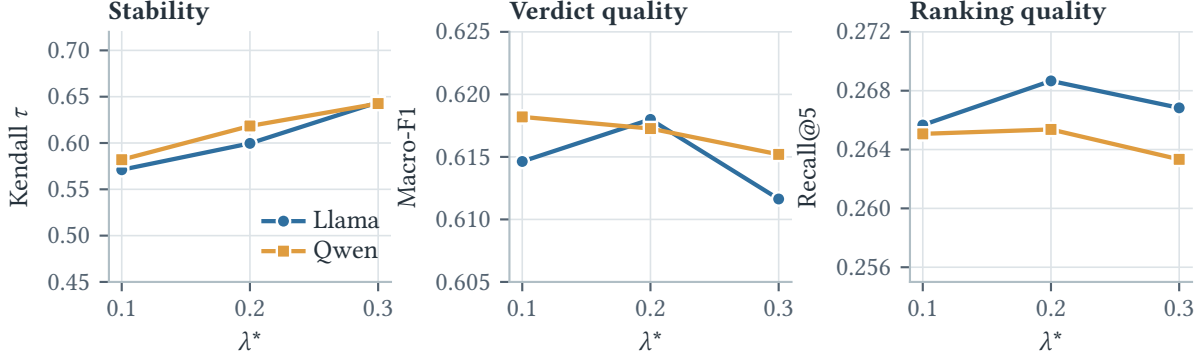


Fig. 17. Effect of the target invariance weight λ^* . Cross-lingual ranking stability rises monotonically with λ^* for both backbones, whereas both task metrics macro-F1 and Recall@5 peak at $\lambda^* = 0.2$. The objective therefore trades a little task quality for consistency beyond that point, making $\lambda^* = 0.2$ the balanced choice.

0.3161 and Fleiss $\kappa = 0.9353$ compared with 0.9195. Mistral-7B is weaker on both dimensions (0.5843 and 0.1197), supporting the decision to keep it as a reference rather than extend it with the invariance objective. No single configuration is best on every measure. *Llama, reg.* has the best English macro-F1 (0.6263) and Recall@5 (0.2710). *Llama, inv. $\lambda^* = 0.2$* has the best Arabic macro-F1 (0.6161), while *Llama, inv. $\lambda^* = 0.3$* has the best ranking stability ($\tau = 0.6440$). This split again shows why accuracy and stability must be considered separately.

5.7 The Relationship Between Invariance and Reasoning Quality (RQ2)

For RQ2, we examine language invariance and reasoning quality as separate properties. This lets us study their relationship directly. Here, *mean macro-F1* represents reasoning quality. It is the claim-verdict macro-F1 averaged across the three languages. We report Recall@5 separately because verdict quality and ranking quality behave differently.

Over the 28 configurations, mean macro-F1 correlates with Kendall τ at $r = 0.60$ and with Fleiss κ at $r = 0.57$. The corresponding correlations for the ranking metric are weaker, mean Recall@5 correlates with τ at only $r = 0.39$ (English Recall@5 alone, $r = 0.49$), while its correlation with κ is $r = 0.57$. Quality and consistency therefore have a positive but moderate relationship, mainly for verdict quality. The amount of relevant evidence in the top five says relatively little about whether the model retrieves the same evidence in another language.

The relationship is not deterministic. Figure 14 shows that configurations with nearly identical mean macro-F1 (0.61–0.62) span τ values from 0.08 to 0.64. Thus, at almost fixed verdict quality, stability varies by more than 0.5. Mean macro-F1 is therefore a weak predictor of stability, and the two properties are separate in practice as well as in principle. This confirms in a ranking setting what prior work found for factual prediction [9, 30], a more accurate multilingual model is not necessarily more consistent, and improving task quality alone does not ensure consistency.

Fleiss κ and Kendall τ correlate at $r = 0.83$ across configurations, so verdict and ranking consistency are related. However, Section 5.3 showed that κ is compressed into a narrow high range while τ varies widely. The measures are therefore not interchangeable, despite their high correlation.

5.8 Threats to Validity

Context-budget confound. The most important limitation concerns the training context. The invariance-trained models used a smaller context budget because aligned trilingual batching multiplies the number of sequences in each optimisation step. The available compute did not support the full long-context setting. Several standard fine-tunes used the full context budget. Training method and context budget are therefore partly entangled, so this is not a fully controlled comparison.

Two observations reduce the concern. First, the confound works against the invariance-trained models, which use the smaller context budget. Their gains are therefore conservative estimates. Second, we can estimate the context effect from architectures tested at both budgets. Moving from the reduced to the full budget changes macro-F1 by +0.0052 in English, +0.0043 in Spanish, and −0.0053 in Arabic. The mean change is only +0.0014, which is smaller than the invariance effect and does not even have the same direction in every language. Under the matched reduced-budget comparison, invariance-trained models still improve mean macro-F1 by +0.0083, Arabic macro-F1 by +0.0197, and Kendall τ by a factor of 4.85. No invariance-trained configuration was tested with the full context budget. That experiment would resolve the confound and is the most important next step. The evidence therefore supports the following guarded claim, *invariance training at a reduced context budget matches or exceeds full-context fine-tuning on mean macro-F1, and dominates it on cross-lingual stability*. The stability result is unlikely to be explained by this confound because the estimated context effect is far too small to produce a fourfold change in τ .

Small comparison arms. The matched-budget comparison contains only two short-context fine-tunes, one Llama and one Qwen. The full-budget group contains seven configurations. The group means are therefore imprecise. Spanish is also a clear exception to the general pattern, *Llama, short context* reaches 0.6220 Spanish macro-F1, above all 13 invariance-trained configurations. We therefore do not claim an improvement for Spanish. The supported conclusion is parity in English and Spanish, an improvement in Arabic, and a large improvement in cross-lingual stability.

Behavioural, not representational, evidence. All invariance measures in this study are behavioural, they compare model outputs across languages. As noted in Section 2.6, matching outputs do not prove that the model stores or processes the underlying knowledge in the same way [17]. The results show that the objective changes what the model *does* across languages, but not what changes inside the model.

5.9 Summary of Findings

RQ1 Does optimising for language-invariant reasoning improve ranking performance? Not on the official ranking metric. Mean Recall@5 is 0.2656 for invariance training against 0.2679 for standard fine-tuning, and the best Recall@5 in the study (0.2710) belongs to a standard fine-tune. What it does produce is a fourfold, fully separated improvement in the cross-lingual stability of the ranking ($\tau = 0.567$ compared with 0.131, $d = 6.09$). It also produces a small but consistent gain in verdict quality (0.6148 compared with 0.5977 mean macro-F1). Per-language ranking quality remains essentially unchanged, but ranking *consistency* across languages improves substantially.

RQ2 What is the relationship between language invariance and reasoning quality? The relationship is positive but moderate. Mean macro-F1, which represents verdict quality across the three languages, correlates with ranking stability at $r = 0.60$ and with verdict agreement at $r = 0.57$. Mean Recall@5 is a weaker predictor, correlating with τ at only $r = 0.39$. Even at almost fixed mean macro-F1, stability varies by more than 0.5 in τ . Consistency is therefore not a by-product of task quality, and the two must be measured separately. The random baseline reinforces this conclusion because it is perfectly invariant but uninformative.

RQ3 Can language-invariant training objectives lead to more robust multilingual reasoning models?

Yes, with the qualifications above. The gains are concentrated where robustness is most needed, in Arabic, which uses a different script and is the least well-represented of the three languages. In Arabic, the gain is concentrated in the minority Conflicting class, where standard fine-tuning degrades several times more than invariance training. The effect appears with both Llama and Qwen, suggesting that it comes from the method rather than one model family. It is also achieved with a reduced training-context budget.

One observation connects all three questions. Standard multilingual fine-tuning already produces models that usually agree on the verdict across languages. It does not produce models that agree on *why*. The invariance objective mainly aligns the evidence, not the final label. Verdict-level agreement cannot reveal this difference, so a standard evaluation protocol would have missed it.

6 Conclusion

This thesis asked whether language invariance is a useful *objective* for multilingual reasoning, rather than only a property worth measuring. We studied multilingual numerical claim verification in CLEF CheckThat! 2026 Task 2 [48]. Given a numerical claim, its evidence, and a set of candidate reasoning traces, a system must rank the traces and derive a claim-level verdict in English, Spanish, and Arabic. We began with the supervised trace scorer from our shared-task submission [22]. We then added a score-level language-invariance objective that penalises different relevance profiles for corresponding traces across languages. We also introduced an evaluation protocol that separates cross-lingual consistency from task accuracy. The study covers 28 valid configurations, three open-weight backbones, and a held-out trilingual test set of 4,233 parallel claims. The results show that invariance is useful, but mainly for cross-lingual robustness rather than per-language ranking quality.

6.1 Answers to the Research Questions

RQ1: Does optimising for language-invariant reasoning improve ranking performance? Not on the official ranking metric, but it makes the ranking much more stable across languages. Invariance training and standard multilingual fine-tuning have indistinguishable Recall@5 scores. The best Recall@5 belongs to a standard fine-tune, and both approaches reach roughly 90% of the metric’s attainable ceiling. However, English–Arabic rank agreement increases roughly fourfold. The stability ranges of the invariance-trained and non-invariance systems do not overlap. Verdict quality also improves modestly. The supported claim is therefore about cross-lingual robustness, not higher ranking accuracy.

The ranking component of this task is already close to saturated. For 13.7% of claims, none of the candidate traces matches the gold verdict. Better ranking cannot improve Recall@5 for those claims. Further gains will require generating or retrieving better candidate traces. Given this ceiling, an objective that preserves retrieval quality while making rankings reproducible across languages improves a dimension that still has substantial room for progress.

RQ2: What is the relationship between language invariance and reasoning quality? The relationship is positive and moderate, but neither property can replace the other. Verdict quality correlates with cross-lingual rank agreement at $r = 0.60$. Yet even at almost fixed verdict quality, the observed agreement spans more than half the range of Kendall’s τ . The ranking metric is an even weaker predictor of consistency. Two findings make this distinction clear. First, the random-ranking baseline is perfectly language-invariant by construction but performs poorly. Consistency alone therefore does not show quality. Second, every evaluated system, including the weakest, has high verdict agreement across languages, while rank agreement varies by almost an order of magnitude. Models often reach the same verdict in all three languages using largely different evidence.

This difference has an important methodological consequence. If we had measured cross-lingual consistency only from final answers, we would have concluded that all systems were already language-invariant. The main

effect would have remained invisible. Agreement on discrete outputs is useful but insufficient. When a system produces an intermediate structure, such as a trace ranking, agreement should also be measured at that level. This extends the known separation between accuracy and consistency from factual prediction to a ranking setting [9, 30].

RQ3: Can language-invariant training objectives lead to more robust multilingual reasoning models? Yes. The gains appear where the multilingual task is most difficult. Spanish behaves almost the same as English for every system, so the main cross-lingual challenge is Arabic. This is consistent with the expectation that Arabic’s different numeral script and weaker representation in pre-training make it harder [36]. Within Arabic, the largest drop occurs in the minority Conflicting class. This label requires the model to weigh contradictory quantitative evidence rather than match a single number. Standard fine-tuning degrades sharply on this class, while invariance training largely maintains performance. The effect appears with both Llama and Qwen, suggesting that it comes from the objective rather than one model family. It was also achieved with a *reduced* training-context budget, making the comparison conservative.

The value-aware numerical components from the shared-task system did not add to this gain. Combining plain-text numeric normalisation with the invariance objective reduced cross-lingual stability and did not improve accuracy. Numeric normalisation still gave a small benefit to standard fine-tunes. The most likely explanation is redundancy rather than general harm. Both methods address differences in how the same quantity is represented across scripts. Once the score-level constraint is active, adding canonical values to the input may increase its length without adding new information. The evidence therefore does not support combining the two methods.

6.2 Contributions Revisited

This thesis delivers the contributions listed in Section 1.5. First, it defines language invariance as a differentiable training objective for a competitive trace-scoring pipeline. The objective works on continuous per-trace scores, which can be aligned directly across languages. Generative rankers produce only discrete orderings that must first be reconciled. Second, the thesis compares ranking formulations from the perspective of consistency. Prompted listwise, pairwise, and preference-based ranking remain sensitive to candidate order, while supervised trace scoring avoids this dependence. Third, it tests value-aware numerical representations together with the invariance objective and reports the negative interaction described above. Finally, it introduces an evaluation protocol that reports task quality and cross-lingual agreement separately at both the verdict and ranking levels. The results show why ranking-level agreement is the more informative measure here.

Two additional findings were not anticipated. First, Recall@5, Precision@5, MRR@5, macro-F1, and the majority-class F1 scores are nearly collinear across configurations. These metrics largely produce the same ordering of systems, while the minority Conflicting class separates them. Second, a seeded random ranker is a useful diagnostic baseline. It is invariant by construction and therefore provides a clear example of why consistency and quality are independent.

6.3 Limitations

Section 5.8 discusses the threats to validity in full. Two limitations are especially important. First, the invariance-trained models used a smaller context budget than several standard fine-tunes, so training method and context budget are partly entangled. This confound is conservative because the invariance-trained group has the smaller budget. The measured effect of extra context is also small and changes direction across languages. Even so, this is an observational comparison rather than a controlled ablation. Second, all invariance measures are behavioural. The results show that the objective changes what a model *does* across languages, but not how it represents the content internally. Matching behaviour does not necessarily imply a shared internal representation [17]. The improvement is also not uniform. The best short-context standard fine-tune has higher Spanish macro-F1 than

every invariance-trained configuration, and the aggregate Arabic gap does not reach statistical significance on its own, although its direction is consistent.

6.4 Future Work

Several directions follow from these results. The most valuable next experiment is also the least expensive, run one invariance-trained configuration with the full context budget. A single seed would provide an initial check and help resolve the context-budget confound. The language set should also be extended to more scripts and language families to test whether the effect generalises beyond Arabic.

A second direction is to study the mechanism. Probing or representational-similarity analysis of the trace scorer’s hidden states [45, 54] could test whether score-level alignment also brings the internal representations of parallel inputs closer. Alternatively, the objective may only constrain the output layer. This distinction would show whether invariance acts as an inductive bias or only as a surface constraint.

A third direction is to address the ranking ceiling. Ranking is nearly saturated on the current candidate set, so future work should generate reasoning traces that cover the gold verdict more often. For the 13.7% of claims with no gold-matching trace, no verifier can improve Recall@5. The score-level invariance objective is also not limited to fact-checking. It can be used wherever a model assigns comparable scores to parallel items. Testing it on other multilingual ranking and reranking tasks would show how widely the result generalises.

6.5 Closing Remarks

The main finding depends on separating two ideas. Standard multilingual fine-tuning already produces models that usually reach the same verdict across languages. It does not ensure that they reach that verdict for the same reasons. Training for language invariance does not improve the model’s ability to find relevant evidence, and it does not measurably change the official ranking metric. Instead, it makes the evidence used by the model more consistent across languages. A claim and its translation are therefore not only assigned the same label, they are more often assessed using the same evidence. In fact-checking, the justification matters as much as the verdict. This property is therefore valuable, but it remains invisible when evaluation measures agreement only on final answers.

References

- [1] Antara Raaghavi Bhattacharya, Isabel Papadimitriou, Kathryn Davidson, and David Alvarez-Melis. 2025. Investigating the interaction of linguistic and mathematical reasoning in language models using multilingual number puzzles. arXiv:2506.13886 [cs.CL] <https://arxiv.org/abs/2506.13886>
- [2] Steven Cao, Nikita Kitaev, and Dan Klein. 2020. Multilingual Alignment of Contextual Word Representations. arXiv:2002.03518 [cs.CL] <https://arxiv.org/abs/2002.03518>
- [3] Yi-Chen Chang, Canasai Kruengkrai, and Junichi Yamagishi. 2023. XFEVER: Exploring Fact Verification across Languages. arXiv:2310.16278 [cs.CL] <https://arxiv.org/abs/2310.16278>
- [4] Primakov Chungkham, V Venkatesh, Vinay Setty, and Avishek Anand. 2025. Think Right, Not More: Test-Time Scaling for Numerical Claim Verification. arXiv:2509.22101 [cs.CL] <https://arxiv.org/abs/2509.22101>
- [5] Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46.
- [6] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised Cross-lingual Representation Learning at Scale. arXiv:1911.02116 [cs.CL] <https://arxiv.org/abs/1911.02116>
- [7] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs. arXiv:2305.14314 [cs.LG] <https://arxiv.org/abs/2305.14314>
- [8] Andreea Dutulescu, Stefan Ruseti, and Mihai Dascalu. 2026. Value-Aware Numerical Representations for Transformer Language Models. arXiv:2601.09706 [cs.CL] <https://arxiv.org/abs/2601.09706>
- [9] Yanai Elazar, Nora Kassner, Shauli Ravfogel, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. 2021. Measuring and Improving Consistency in Pretrained Language Models. arXiv:2102.01017 [cs.CL] <https://arxiv.org/abs/2102.01017>

- [10] Constanza Fierro and Anders Søgaard. 2022. Factual Consistency of Multilingual Pretrained Language Models. In *Findings of the Association for Computational Linguistics: ACL 2022*. Association for Computational Linguistics, 3046–3052. doi:10.18653/v1/2022.findings-acl.240
- [11] Joseph L. Fleiss. 1971. Measuring Nominal Scale Agreement among Many Raters. *Psychological Bulletin* 76, 5 (1971), 378–382.
- [12] Xiyang Fu and Wei Liu. 2025. How Reliable is Multilingual LLM-as-a-Judge? arXiv:2505.12201 [cs.CL] <https://arxiv.org/abs/2505.12201>
- [13] Ashim Gupta and Vivek Srikumar. 2021. X-FACT: A New Benchmark Dataset for Multilingual Fact Checking. arXiv:2106.09248 [cs.CL] <https://arxiv.org/abs/2106.09248>
- [14] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685 [cs.CL] <https://arxiv.org/abs/2106.09685>
- [15] Kaiyu Huang, Fengran Mo, Xinyu Zhang, Hongliang Li, You Li, Yuanchi Zhang, Weijian Yi, Yulong Mao, Jinchun Liu, Yuzhuang Xu, et al. 2024. A Survey on Large Language Models with Multilingualism: Recent Advances and New Frontiers. *arXiv preprint arXiv:2405.10936* (2024).
- [16] Katharina Hämmerl, Jindřich Libovický, and Alexander Fraser. 2024. Understanding Cross-Lingual Alignment – A Survey. arXiv:2404.06228 [cs.CL] <https://arxiv.org/abs/2404.06228>
- [17] Maxim Ifergan, Leshem Choshen, Roei Aharoni, Idan Szpektor, and Omri Abend. 2024. Beneath the Surface of Consistency: Exploring Cross-lingual Knowledge Representation Sharing in LLMs. arXiv:2408.10646 [cs.CL] <https://arxiv.org/abs/2408.10646>
- [18] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Léo Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7B. arXiv:2310.06825 [cs.CL] <https://arxiv.org/abs/2310.06825>
- [19] Zhengbao Jiang, Antonios Anastasopoulos, Jun Araki, Haibo Ding, and Graham Neubig. 2020. X-FACTR: Multilingual Factual Knowledge Retrieval from Pretrained Language Models. arXiv:2010.06189 [cs.CL] <https://arxiv.org/abs/2010.06189>
- [20] Maurice G. Kendall. 1938. A New Measure of Rank Correlation. *Biometrika* 30, 1/2 (1938), 81–93.
- [21] Ahmad Raza Khawaja. 2026. Numerical Fact-Checking: Code, Configurations and Evaluation Scripts. <https://github.com/ahmadrazakhawaja/numerical-fact-checking>. Accessed: 2026-08-15.
- [22] Ahmad Raza Khawaja and Adam Moshe Lehari. 2026. GippLab at CheckThat! 2026: Multilingual Numerical Claim Verification with QLoRA Trace Scoring, See [38].
- [23] Klaus Krippendorff. 2004. *Content Analysis: An Introduction to Its Methodology* (2nd ed.). Sage Publications.
- [24] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. 2022. Solving Quantitative Reasoning Problems with Language Models. arXiv:2206.14858 [cs.CL] <https://arxiv.org/abs/2206.14858>
- [25] Tianyu Liu, Jirui Qi, Mrinmaya Sachan, Ryan Cotterell, Raquel Fernández, and Arianna Bisazza. 2026. Post-Training Language Models for Crosslingual Consistency. arXiv:2603.04678 [cs.CL] <https://arxiv.org/abs/2603.04678>
- [26] Tianqi Liu, Zhen Qin, Junru Wu, Jiaming Shen, Misha Khalman, Rishabh Joshi, Yao Zhao, Mohammad Saleh, Simon Baumgartner, Jialu Liu, Peter J Liu, and Xuanhui Wang. 2025. LIPO: Listwise Preference Optimization through Learning-to-Rank. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Luis Chiruzzo, Alan Ritter, and Lu Wang (Eds.). Association for Computational Linguistics, Albuquerque, New Mexico, 2404–2420. doi:10.18653/v1/2025.naacl-long.121
- [27] Meta AI. 2024. Meta Llama 3.1 8B Instruct. <https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct> Accessed: 2026-05-17.
- [28] Swaroop Mishra, Arindam Mitra, Neeraj Varshney, Bhavdeep Sachdeva, Peter Clark, Chitta Baral, and Ashwin Kalyan. 2022. NumGLUE: A Suite of Fundamental yet Challenging Mathematical Reasoning Tasks. arXiv:2204.05660 [cs.CL] <https://arxiv.org/abs/2204.05660>
- [29] Telmo Pires, Eva Schlinger, and Dan Garrette. 2019. How multilingual is Multilingual BERT? arXiv:1906.01502 [cs.CL] <https://arxiv.org/abs/1906.01502>
- [30] Jirui Qi, Raquel Fernández, and Arianna Bisazza. 2023. Cross-Lingual Consistency of Factual Knowledge in Multilingual Language Models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 10650–10666. doi:10.18653/v1/2023.emnlp-main.658
- [31] Jingfen Qiao, Jin Huang, Xinyu Ma, Shuaiqiang Wang, Dawei Yin, Evangelos Kanoulas, and Andrew Yates. 2026. LLM-based Listwise Reranking under the Effect of Positional Bias. arXiv:2604.03642 [cs.IR] <https://arxiv.org/abs/2604.03642>
- [32] Zhen Qin, Rolf Jagerman, Kai Hui, Honglei Zhuang, Junru Wu, Le Yan, Jiaming Shen, Tianqi Liu, Jialu Liu, Donald Metzler, Xuanhui Wang, and Michael Bendersky. 2024. Large Language Models are Effective Text Rankers with Pairwise Ranking Prompting. arXiv:2306.17563 [cs.IR] <https://arxiv.org/abs/2306.17563>
- [33] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2024. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. arXiv:2305.18290 [cs.LG] <https://arxiv.org/abs/2305.18290>
- [34] Roussel Rahman. 2025. Large Language Models in Numberland: A Quick Test of Their Numerical Reasoning Abilities. arXiv:2504.00226 [cs.AI] <https://arxiv.org/abs/2504.00226>

- [35] Roussel Rahman and Aashwin Ananda Mishra. 2025. A Fragile Number Sense: Probing the Elemental Limits of Numerical Reasoning in LLMs. arXiv:2509.06332 [cs.LG] <https://arxiv.org/abs/2509.06332>
- [36] Varshini Reddy, Craig W. Schmidt, Seth Ebner, Adam Wiemerslage, Yuval Pinter, and Chris Tanner. 2026. The Effect of Scripts and Formats on LLM Numeracy. arXiv:2601.15251 [cs.CL] <https://arxiv.org/abs/2601.15251>
- [37] Premtim Sahitaj, Iffat Maab, Junichi Yamagishi, Jawan Kolanowski, Sebastian Möller, and Vera Schmitt. 2025. Towards Automated Fact-Checking of Real-World Claims: Exploring Task Formulation and Assessment with LLMs. arXiv:2502.08909 [cs.CL] <https://arxiv.org/abs/2502.08909>
- [38] Eva Sánchez Salido, Alberto Barrón-Cedeño, Alba García Seco de Herrera, Sean MacAvaney, and Julia Maria Struß (Eds.). 2026. *CLEF 2026 Working Notes*. Jena, Germany.
- [39] Lisa Schut, Yarin Gal, and Sebastian Farquhar. 2025. Do Multilingual LLMs Think In English? arXiv:2502.15603 [cs.CL] <https://arxiv.org/abs/2502.15603>
- [40] Eli Schwartz, Leshem Choshen, Joseph Shtok, Sivan Doveh, Leonid Karlinsky, and Assaf Arbelle. 2024. NumeroLogic: Number Encoding for Enhanced LLMs’ Numerical Reasoning. arXiv:2404.00459 [cs.CL] <https://arxiv.org/abs/2404.00459>
- [41] Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. Neural Machine Translation of Rare Words with Subword Units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Berlin, Germany, 1715–1725. doi:10.18653/v1/P16-1162
- [42] Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2022. Language Models are Multilingual Chain-of-Thought Reasoners. arXiv:2210.03057 [cs.CL] <https://arxiv.org/abs/2210.03057>
- [43] Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. 2025. Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. arXiv:2406.07791 [cs.CL] <https://arxiv.org/abs/2406.07791>
- [44] Charles Spearman. 1904. The Proof and Measurement of Association between Two Things. *The American Journal of Psychology* 15, 1 (1904), 72–101.
- [45] Chris M. Stolle, Rongjun Yu, and Yi Huang. 2026. The reasoning-like capabilities of large language models across different languages: Insights from representational similarity analysis. *Computers in Human Behavior: Artificial Humans* 7 (2026), 100250. doi:10.1016/j.chbah.2026.100250
- [46] Qwen Team. 2025. Qwen3 Technical Report. arXiv:2505.09388 [cs.CL] <https://arxiv.org/abs/2505.09388>
- [47] Venkatesh V. Abhijit Anand, Avishek Anand, and Vinay Setty. 2024. QuanTemp: A real-world open-domain benchmark for fact-checking numerical claims. arXiv:2403.17169 [cs.CL] <https://arxiv.org/abs/2403.17169>
- [48] Venkatesh V. Vinay Setty, Primakov Chungkham, and Avishek Anand. 2026. Overview of the CLEF-2026 CheckThat! Lab Task 2 on Fact-Checking Numerical Claims, See [38].
- [49] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is All you Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30. <https://arxiv.org/abs/1706.03762>
- [50] J. Wahle, T. Ruas, S. M. Mohammad, N. Meuschke, and B. Gipp. 2023. AI Usage Cards: Responsibly Reporting AI-Generated Content. In *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE Computer Society, Los Alamitos, CA, USA, 282–284. doi:10.1109/JCDL57899.2023.00060
- [51] Eric Wallace, Yizhong Wang, Sujian Li, Sameer Singh, and Matt Gardner. 2019. Do NLP Models Know Numbers? Probing Numeracy in Embeddings. arXiv:1909.07940 [cs.CL] <https://arxiv.org/abs/1909.07940>
- [52] Yumeng Wang, Zhiyuan Fan, Qingyun Wang, May Fung, and Heng Ji. 2025. CALM: Unleashing the Cross-Lingual Self-Aligning Ability of Language Model Question Answering. arXiv:2501.18457 [cs.CL] <https://arxiv.org/abs/2501.18457>
- [53] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. <http://arxiv.org/abs/2201.11903> tex.ids= WeiWSB23a, WeiWSB23b arXiv: 2201.11903 [cs].
- [54] Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do Llamas Work in English? On the Latent Language of Multilingual Transformers. arXiv:2402.10588 [cs.CL] <https://arxiv.org/abs/2402.10588>
- [55] Shijie Wu, Alexis Conneau, Haoran Li, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Emerging Cross-lingual Structure in Pretrained Language Models. arXiv:1911.01464 [cs.CL] <https://arxiv.org/abs/1911.01464>
- [56] Bo Zheng, Li Dong, Shaohan Huang, Wenhui Wang, Zewen Chi, Saksham Singhal, Wanxiang Che, Ting Liu, Xia Song, and Furu Wei. 2021. Consistency Regularization for Cross-Lingual Fine-Tuning. arXiv:2106.08226 [cs.CL] <https://arxiv.org/abs/2106.08226>

A Full Per-Model Results

Tables 3 and 4 give the complete per-model figures condensed in Table 2. All values are means over three random seeds on the aligned trilingual test set of 4,233 parallel claims. The three invalid configurations described in Section 5.1 are omitted. The random baseline is also omitted here because it is reported only as a floor in Table 2 and excluded from the statistical analysis.

Model	Recall@5			Macro-F1			
	EN	ES	AR	EN	ES	AR	Mean
<i>Invariance-trained (this thesis)</i>							
Qwen, inv. $\lambda^*=0.1$	0.2663	0.2667	0.2622	0.6192	0.6197	0.6157	0.6182
Llama, inv. $\lambda^*=0.2$	0.2699	0.2694	0.2667	0.6211	0.6168	0.6161	0.6180
Qwen, inv. $\lambda^*=0.2$	0.2661	0.2670	0.2630	0.6176	0.6185	0.6157	0.6173
Llama, inv. + num. norm.	0.2697	0.2698	0.2618	0.6227	0.6205	0.6052	0.6161
Qwen, inv. $\lambda^*=0.05$	0.2659	0.2663	0.2627	0.6174	0.6186	0.6115	0.6158
Qwen, inv. $\lambda^*=0.3$	0.2646	0.2648	0.2606	0.6173	0.6185	0.6098	0.6152
Qwen, inv. + reg. + num. norm.	0.2648	0.2671	0.2552	0.6211	0.6206	0.6037	0.6151
Qwen, inv. + reg.	0.2677	0.2657	0.2627	0.6202	0.6158	0.6090	0.6150
Llama, inv. $\lambda^*=0.1$	0.2671	0.2681	0.2618	0.6170	0.6175	0.6094	0.6146
Llama, inv. + reg.	0.2692	0.2693	0.2627	0.6177	0.6132	0.6089	0.6133
Llama, inv. $\lambda^*=0.3$	0.2683	0.2684	0.2638	0.6165	0.6150	0.6034	0.6116
Llama, inv. + reg. + num. norm.	0.2689	0.2706	0.2620	0.6137	0.6190	0.6018	0.6115
Qwen, inv. + num. norm.	0.2663	0.2662	0.2574	0.6163	0.6146	0.6003	0.6104
<i>Standard multilingual fine-tuning</i>							
Llama, standard	0.2649	0.2704	0.2692	0.6256	0.6320	0.6105	0.6227
Llama, reg.	0.2710	0.2783	0.2673	0.6263	0.6297	0.5808	0.6123
Qwen, num. norm.	0.2702	0.2762	0.2654	0.6221	0.6258	0.5810	0.6096
Qwen, short context	0.2642	0.2672	0.2649	0.6101	0.6161	0.6015	0.6092
Llama, num. norm.	0.2640	0.2732	0.2664	0.6154	0.6226	0.5854	0.6078
Llama, reg. + num. norm.	0.2583	0.2723	0.2649	0.6072	0.6261	0.5825	0.6053
Llama, short context	0.2632	0.2711	0.2583	0.6134	0.6220	0.5761	0.6038
Qwen, standard	0.2671	0.2737	0.2656	0.6135	0.6203	0.5715	0.6018
Llama, BCE head	0.2684	0.2729	0.2654	0.6085	0.6066	0.5728	0.5960
<i>Earlier trace scorers</i>							
Mistral, multiling. trace scorer (10 ep.)	0.2566	0.2549	0.2520	0.5967	0.5932	0.5828	0.5909
Mistral, trace scorer	0.2196	0.2490	0.2171	0.5847	0.5971	0.5775	0.5864
Mistral, multiling. trace scorer	0.2140	0.2430	0.2185	0.5806	0.5915	0.5864	0.5862
Mistral, numeric trace scorer	0.2184	0.2155	0.2228	0.5826	0.5693	0.5938	0.5819
<i>Prompted baselines</i>							
Mistral, prompted (zero-shot)	0.2225	0.2173	0.2129	0.5771	0.5795	0.5718	0.5761
Llama, prompted (zero-shot)	0.2234	0.2222	0.2219	0.5751	0.5716	0.5792	0.5753
<i>Reported for reference only, excluded from every statistic</i>							
Random ranking (floor)	0.2181	0.2181	0.2181	0.5820	0.5820	0.5820	0.5820
Llama, inv. $\lambda^*=0.05^\dagger$	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938
Mistral, English DPO [†]	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938
Qwen, prompted (zero-shot) [†]	0.2228	0.2228	0.2228	0.5938	0.5938	0.5938	0.5938
Recall@5 upper bound	0.3005			—			

Table 3. Per-model results on the held-out trilingual test set. *Mean* averages over the three languages; best value per column in bold, over the 28 valid configurations. All values are means over three random seeds. Rows marked [†] are excluded from every statistic: those runs degenerated to emitting the input order rather than ranking (Section 5.1).

Model	Verdict agreement		Ranking agreement (Kendall τ)		
	Complete	Fleiss κ	EN-ES	EN-AR	ES-AR
<i>Invariance-trained (this thesis)</i>					
Llama, inv. $\lambda^*=0.3$	0.9631	0.9428	0.8268	0.6440	0.6369
Qwen, inv. $\lambda^*=0.3$	0.9754	0.9623	0.8226	0.6426	0.6354
Qwen, inv. $\lambda^*=0.2$	0.9655	0.9470	0.7922	0.6184	0.6023
Llama, inv. $\lambda^*=0.2$	0.9657	0.9471	0.8316	0.5996	0.5941
Qwen, inv. $\lambda^*=0.1$	0.9648	0.9479	0.7680	0.5820	0.5671
Qwen, inv. + reg.	0.9627	0.9424	0.7396	0.5720	0.5411
Llama, inv. $\lambda^*=0.1$	0.9624	0.9425	0.7909	0.5711	0.5527
Llama, inv. + reg.	0.9634	0.9441	0.7830	0.5597	0.5472
Qwen, inv. $\lambda^*=0.05$	0.9646	0.9460	0.7274	0.5294	0.5111
Qwen, inv. + num. norm.	0.9622	0.9420	0.7479	0.5188	0.5138
Qwen, inv. + reg. + num. norm.	0.9587	0.9379	0.7473	0.5115	0.5060
Llama, inv. + reg. + num. norm.	0.9617	0.9394	0.7762	0.5093	0.4944
Llama, inv. + num. norm.	0.9579	0.9347	0.7666	0.5089	0.4998
<i>Standard multilingual fine-tuning</i>					
Qwen, num. norm.	0.9549	0.9272	0.3106	0.2496	0.2210
Qwen, standard	0.9407	0.9053	0.2683	0.1696	0.1496
Llama, reg. + num. norm.	0.9287	0.8872	0.1976	0.1261	0.1218
Llama, short context	0.9185	0.8738	0.2377	0.1232	0.1012
Llama, BCE head	0.9707	0.9519	0.1245	0.1159	0.1293
Llama, reg.	0.9431	0.9095	0.1912	0.1117	0.1146
Qwen, short context	0.9320	0.8948	0.1628	0.1107	0.0672
Llama, standard	0.9206	0.8810	0.2079	0.0863	0.1010
Llama, num. norm.	0.9341	0.8949	0.1732	0.0838	0.1456
<i>Earlier trace scorers</i>					
Mistral, numeric trace scorer	0.9022	0.8643	0.0435	0.1075	0.0933
Mistral, multiling. trace scorer (10 ep.)	0.9398	0.9040	0.0886	0.0955	0.0855
Mistral, multiling. trace scorer	0.8802	0.8298	0.0181	0.0161	0.0137
Mistral, trace scorer	0.8885	0.8404	-0.0084	0.0122	0.0179
<i>Prompted baselines</i>					
Mistral, prompted (zero-shot)	0.9395	0.9168	0.6754	0.3674	0.4023
Llama, prompted (zero-shot)	0.9320	0.9049	0.1242	0.0697	0.3416
<i>Reported for reference only, excluded from every statistic</i>					
Random ranking (floor)	—	—	—	—	—
Llama, inv. $\lambda^*=0.05^\dagger$	—	—	—	—	—
Mistral, English DPO [†]	—	—	—	—	—
Qwen, prompted (zero-shot) [†]	—	—	—	—	—

Table 4. Cross-lingual consistency, measured separately from task accuracy. *Complete* is the fraction of claims on which all three languages produce the same verdict; ranking agreement is the mean pairwise Kendall τ between the language-conditioned trace orderings. Best per column in bold. All agreement values are dashed (—) for the random ranking and for the invalid runs ([†]): each of them scores exactly 1.0000 on every measure, the random ranking because it ignores the input by construction and the others because the same predictions were compared with themselves, so the figure reflects the artefact rather than any property of the system.

B Prompt Templates

This section reproduces the prompt templates used by the two generative ranking baselines (Section 4.7) and the input template used by the supervised trace scorer (Section 4.3). Braces denote fields substituted at run time. All prompted baselines additionally use the system message

You are a careful multilingual fact-checking assistant. Return strict JSON only.
and are decoded greedily, so the only source of variation between runs is the input itself.

You are ranking candidate reasoning traces for multilingual fact-checking.
Use the claim and the evidence snippets to judge which reasoning traces are most reliable.
Rank all candidate traces from best to worst. The input order is arbitrary and must not be copied unless it is truly the best ranking.
A better trace is one that is better supported by the evidence, checks the claim more directly, and reaches a more reliable verdict.

There are {num_traces} traces indexed 0 through {num_traces-1}.
The ranked_trace_indices value must be a permutation containing each of these indices exactly once: [{valid_indices}].
Do not omit indices. Do not repeat indices. Do not use placeholder indices.
Then predict the final verdict for the claim.

Allowed verdict labels: {allowed_labels}

Return strict JSON only with this schema:
{"ranked_trace_indices":[...],"predicted_verdict":"<one allowed label>"}

Do not include markdown. Do not include any explanation.

Claim:
{claim}

Evidence snippets:
[0] {evidence_0} . . .

Candidate reasoning traces:
[0] {trace_0} . . .

Table 5. Zero-shot **listwise** ranking prompt. The model receives all candidate traces at once and must return a complete permutation of trace indices together with a claim verdict. The explicit instruction not to copy the input order is included precisely because listwise rankers are prone to reproducing it (Section 5.2).

You are comparing two candidate reasoning traces for multilingual fact-checking.
 Use the claim and evidence snippets to decide which trace is more reliable.
 Choose exactly one trace: A or B.

Return strict JSON only in this exact schema:

```
{"preferred_trace": "A"}
```

Allowed values for preferred_trace: A, B.

Do not include markdown. Do not include any explanation.

Claim:

```
{claim}
```

Evidence snippets:

```
[0] {evidence_0} . . .
```

Trace A (original index {left_index}):

```
{left_trace}
```

Trace B (original index {right_index}):

```
{right_trace}
```

Table 6. Zero-shot **pairwise** ranking prompt. Each claim requires $n(n-1)/2$ calls in the balanced orientation and $n(n-1)$ in the bidirectional orientation; the ranking is assembled from the accumulated wins.

Claim: {claim}

Evidence snippets:

```
[0] {evidence_0}
```

```
[1] {evidence_1} . . .
```

Normalized numbers:

(optional)

Claim: {surface}={canonical}, . . .

Evidence: {surface}={canonical}, . . .

Trace: {surface}={canonical}, . . .

Verdict: {trace_verdict}

Justification: {cleaned_trace}

Table 7. Input template for the supervised **trace scorer**. Unlike the prompted baselines this is not an instruction: it is the sequence classified by the model, which emits a single relevance score per claim–trace pair. The trace justification is stripped of any restated verdict so that the scorer cannot read the label off the text, and the optional Normalized numbers block is present only in the numeric-normalization configurations (Section 4.6).

C Technical Information

All experiments were run on NVIDIA A100 GPUs on the institutional cluster, with distributed data-parallel training across two GPUs via `torchrun` for the main trace-scorer runs and a single GPU for inference and smaller ablations. Table 8 lists the software environment. The training, inference and evaluation code, together with the configuration files for every reported run, is available at <https://github.com/ahmadrazakhawaja/numerical-fact-checking> [21]. Together with the versions in Table 8, it is intended to make the reported results reproducible: model behaviour under quantized fine-tuning is sensitive to the exact versions of transformers, peft, and bitsandbytes in particular.

Library	Version
Python	3.12.8
CUDA	12.8
PyTorch	2.10.0
transformers	5.2.0
tokenizers	0.22.2
peft	0.18.1
accelerate	1.12.0
bitsandbytes	0.49.2
safetensors	0.7.0
huggingface_hub	1.4.1
sentencepiece	0.2.1
NumPy	2.4.2
googletrans	4.0.2
tqdm	4.67.3

Table 8. Software environment used for the experiments. Specifying exact versions supports reproducibility and independent validation of the reported results.

Backbone	Parameters	Checkpoint
Mistral-7B-Instruct-v0.3	7B	mistralai/Mistral-7B-Instruct-v0.3
Llama-3.1-8B-Instruct	8B	meta-llama/Llama-3.1-8B-Instruct
Qwen3-8B	8B	Qwen/Qwen3-8B

Table 9. Backbone models. All three are open-weight instruction-tuned checkpoints obtained from the Hugging Face Hub and adapted with QLoRA (Section 4.3).

D AI Usage Card

Generative AI use in preparing this thesis is documented below, following the AI Usage Card standard.

AI Usage Card			
PROJECT DETAILS	PROJECT NAME Language-Invariant Reason Ranking in Numerical Fact-Checking	DOMAIN Natural Language Processing; automated fact-checking	KEY APPLICATION Multilingual numerical claim verification and trace ranking
CONTACT(S)	NAME(S) Ahmad Raza Khawaja	EMAIL(S) ahmad.khawaja@stud.uni-goettingen.de	AFFILIATION(S) University of Göttingen
MODEL(S)	MODEL NAME(S) GPT-5.5		
LITERATURE REVIEW	FINDING LITERATURE Suggesting related work, all suggestions were verified.	FINDING EXAMPLES FROM KNOWN LITERATURE OR ADDING LITERATURE FOR EXISTING STATEMENTS Used for Finding examples from the related literature.	COMPARING LITERATURE
WRITING	GENERATING NEW TEXT BASED ON INSTRUCTIONS	ASSISTING IN IMPROVING OWN CONTENT OR PARAPHRASING RELATED WORK Condensing and copy editing human written own text.	PUTTING OTHER WORKS IN PERSPECTIVE
CODING	GENERATING NEW CODE BASED ON DESCRIPTIONS OR EXISTING CODE Analysis and plotting scripts for the reported statistics, figures, and tables.	REFACTORING AND OPTIMIZING EXISTING CODE Used to refactor and check for bugs and quality improvement of the code written by the author himself.	COMPARING ASPECTS OF EXISTING CODE
ETHICS	WHY DID WE USE AI FOR THIS PROJECT? To speed up drafting, analysis, and typesetting, I as an author take full responsibility for all claims.	WHAT STEPS ARE WE TAKING TO MITIGATE ERRORS OF AI? All content generated from AI is reviewed thoroughly to ensure there are no errors, or instance of hallucination.	WHAT STEPS ARE WE TAKING TO MINIMIZE THE CHANCE OF HARM OR INAPPROPRIATE USE OF AI? Multiple rigorous manual review cycles were done to ensure no content which has no backing is passed through.
THE CORRESPONDING AUTHORS VERIFY AND AGREE WITH THE MODIFICATIONS OR GENERATIONS OF THEIR USED AI-GENERATED CONTENT			
AI Usage Card v2.6		https://ai-cards.org	[50]